

MACROSCOPIC ACTIVITY-BASED MODELING OF URBAN ACTIVE MOBILITY

Romain Azaïs[†], Adrien Marion[†], and Florian Patout[‡]

[†] Inria Lyon, France [‡] INRAE Avignon, France

Abstract. This paper develops a macroscopic, activity-based model of urban active mobility using non-intrusive sensor data. It introduces attendance functions to describe spatio-temporal travel patterns between activities and formulates the disaggregation of aggregated counts as a statistical inference problem. Counts are modeled as Poisson variables, and unknown sub-population sizes are estimated via maximum likelihood, with theoretical guarantees and an efficient EM algorithm for computation. Grounded in a microscopic stochastic model, the framework provides a scalable and privacy-preserving approach to analyzing urban soft mobility dynamics.

1 Introduction

Today’s cities face rapid growth and pressing environmental challenges. To achieve the transition toward more sustainable urban environments, transportation plays a pivotal role, as promoted by the United Nations’11th Sustainable Development Goal¹. The drive to qualify and quantify urban transport sustainability – and to identify the efforts required to achieve it – is fundamental in shaping and supporting public policy [1, 2]. A precise understanding of urban mobility dynamics is particularly essential to this effort [3, 12, 14].

In this article, we examine travel patterns in public spaces, specifically for soft mobility modes such as walking and cycling. Although individual trips are inherently unpredictable, collective patterns emerge when journeys between two different activities – such as shopping, leisure and work – are regrouped together [20]. The aim of this article is to propose a deterministic model that captures the dynamics of each group. It is derived from the large-population limit of a microscopic, individual-based stochastic model. Combined with street-level movement observations, it will offer a precise characterization of expected travel dynamics.

Traditionally, descriptive studies on mobility have relied on surveys [7]. In France, EMC² surveys are conducted in various urban areas across the country [17], overseen by Cerema. They provide a snapshot of resident’s travel patterns over the course of a typical day at a macroscopic scale (district, city), taking into account the mode of transportation used, the purpose of the trip, age, gender, and other factors. Conducting such surveys is resource-intensive. For example, the ongoing survey in the Lyon metropolitan area is expected to take several years and cost €2.4 million². Additionally, investigators face growing challenges in reaching or recruiting households, due to both declining public interest and logistical barriers – such as the lack of landline connections, intercom systems at building entrances, and other access-related issues [17].

Another type of studies make use of the global system for mobile communications (GSM) data collected by carriers to reconstruct urban travel patterns [4]. Various methods for analyzing this data have been developed to complement field studies [10, 22]. However, they raise significant issues, particularly regarding

¹<https://docs.un.org/en/A/RES/71/313>, last consulted on April 12, 2026

²<https://www.francemarches.com/files/BOAMP/25-70604.html>, last consulted on April 12, 2026

anonymization and selection bias [13, 18]. GPS data constitute an additional source of tracking information. These are more accurate than data from cell towers, but at the cost of stricter anonymization constraints. European studies conducted to date have been carried out, but only on a limited panel of volunteers over periods not exceeding a few months [9, 19].

In this article, we propose a complementary approach to these existing methods, focusing on automatic urban counts from sensors. This data is non-intrusive, free from selection bias, openly accessible, and updated in real time [15, 23]. It is highly reliable, with minimal gaps in the time-series records. Automatic counters are becoming widespread in major cities, with accelerated deployment in recent years: in Lyon, for example, 184 bike counters and 72 pedestrian counters are now operational, including 40 installed since 2021³. These devices capture local traffic counts every hour for the main soft modes of transport (pedestrians, electric scooters, bicycles). However, their main drawback is that these counts aggregate passages without distinguishing their link with activities.

We transform the challenge of disaggregating the count data into a statistical estimation problem by introducing a dynamical model for each type of journey, based on additional geographic data of two types: the spatial distribution of activity locations (distribution of shops, residential areas, parks, etc.) and activity schedules (hours of work, frequentation of shops, etc. [11]). Using these data, we model the expected dynamics for each journey type over the study area and within a fixed time interval by designing trips for each pair of activities. This model captures hourly and location-specific movement trends, represented by an *attendance function*. This function is by definition adimensional, akin to a frequency function. It provides a spatio-temporal mapping that encapsulates all the travel information for a given journey type. It will be a key insight of our study.

Each attendance function is associated with a scale parameter. It represents the size of the corresponding subpopulation, i.e. the number of trips made across the study area and time interval for the given journey type. Combined together, the attendance function and the vector of subpopulation sizes give an expected number of passages at each location and each time step. To disaggregate the hidden counts, our strategy is thus to infer these scaling parameters, represented as a vector whose dimension corresponds to the number of journey types.

To that end, we develop a statistical framework that links the count data to the previous model. Count data are modeled as a noisy version of the expected number of passages. This yields on an analytical expression for the likelihood involving the attendance function and the count data, depending on the unknown vector of subpopulation sizes. This vector is also inferred using the Maximum Likelihood Estimator (MLE). The efficiency of this estimator improves with increasing data volume, and it is possible to quantify the associated uncertainties. In terms of asymptotics, several approaches are possible: reducing the counting time period, concatenating data over multiple days (or other cycle of a repeated phenomenon), or increasing the total number of counters. We will focus on the latter option, as the observation period is inherently constrained by the counting devices themselves. Aggregating data over multiple days would undermine the value of real-time analysis and, moreover, introduces challenges in identifying comparable or homogeneous days.

The maximum likelihood estimator we study does not admit a closed form solution. To compute it, we adapt the Expectation-Maximization (EM) algorithm, which is particularly well-suited to our context of hidden data. The inputs are the count data and the attendance functions, evaluated at the counter's locations. Its theoretical efficiency is matched by the rapid execution of the expectation and maximization steps in our context. Finally, this model provides precise insight into the dynamics of soft modes of transportation across the entire area for each activity, all while relying solely on non-invasive data.

The rest of this article is organized as follows. In Section 2 we link counting data to attendance functions and to the unknown vector of subpopulation sizes. Together, these objects allow us to construct expected passages in every point and at every time, for each type of journey, that should be close to the hidden data. As the expected passages depend on the unknown vector of subpopulation sizes, Section 3 introduces the

³<https://data.grandlyon.com/portail/fr/jeux-de-donnees/sites-comptage-metropole-lyon/info>, last consulted on April 12 2026

statistical tools to estimate this parameter using the maximum likelihood method, and presents asymptotic results on its accuracy. In Section 4 we propose an adaptation of the EM algorithm to compute the MLE, addressing the computational challenges of the estimation process. Finally we provide a precise way to construct and interpret the attendance functions in the unidimensional case, by studying a microscopic, individual-based stochastic model in large population. Throughout the article, we illustrate our results using a guiding example, whose expert static data and count data are generated *in silico*. Appendix A provides detailed explanations of the classical results for the MLE, presenting the classical results that we adapt to fit our specific framework. It also includes the computations for the expectation and maximisation steps of our algorithm presented in Section 4. Finally Appendix C contains proofs of formulas presented in Section 5.

2 Setting of the problem

Our strategy to study travel dynamics is based on data from automatic, non intrusive, counters. As they only capture localized snapshots that are aggregated across all journey types, we need a method to disaggregate them and link the counts to activities. We introduce the mathematical tools and notations for our model. In this section, we describe the outputs of the counters and clarify the hidden data.

The urban area is a compact metric set $\mathcal{E}$, such as a segment, a subset of $\mathbb{R}^2$, or a graph, that may represent a street, a field, or a portion of a map. Trips are of K distinct types, each representing a journey between two different activities (leisure, shopping, work, staying home, etc). For instance, commuting trips are treated as two distinct types of journeys: home to work and work to home; the activities involved are in both “staying at home” and “going to work”.

We assume that all individuals are moving without turning back during their trip. This ensures that they are counted at most once by a counter. Inside a fixed time-interval, the counters record the number of individuals passing through their position $x \in \mathcal{E}$. The time step of their data is rough, commonly one hour; we will use the index i for it. Therefore, the number of passages recorder over the time i by the counter at position x is denoted by $n(i, x)$. Additionally, we define the hidden data $n_k(i, x)$ as the number of passages associated with journey type k at the same counter. This hidden data is not directly accessible to this counter, which only records the aggregate sum of all passages,

$$n(i, x) = \sum_{k=1}^K n_k(i, x). \quad (1)$$

We next show how to link this count and the associated unknown vector of sub-population sizes to an attendance function that we define accordingly. This transforms the disaggregation challenge into a problem of estimating the vector of population sizes.

2.1 From a disaggregation challenge to a vector estimation problem

We introduce the notation N for the total number of trips undertaken during the time period of interest $1 \leq i \leq I$ and within the spatial domain $\mathcal{E}$. A trip is here defined as the path an individual takes from an origin activity to a destination activity.

As we have hidden data, we need to categorize the total number per activity, i.e. $N = (N_k)_{k \leq K}$, where each N_k corresponds to journey of type k . For instance, if the time step is one hour and $I = 24$, N is the vector whose coordinate N_k represents the total number of trips corresponding to the journey type k , throughout the day and across the entire study area $\mathcal{E}$.

It should be noted that there is no straightforward relationship linking the count data $n(i, x)$ to the vector N : individuals may be counted by multiple counters during their journeys. Furthermore, it is not guaranteed that all individuals are counted at least once by any counter.

With the introduction of N , we can thus now interpret $n_k(i, x)$, the count of passages of type k , as a sample of all the trips traveled over $\mathcal{E}$, of this type k . Of course, the difficulty is that N_k is unknown: it should be regarded as oracle data, inaccessible through any field surveys. Furthermore, $n_k(i, x)$ is not directly accessible from the counter.

To solve these issues, we focus on the *empirical attendance*, defined as the ratio

$$A_k(i, x) := \frac{n_k(i, x)}{N_k}. \quad (2)$$

This dimensionless function accounts for the empirical probability that a trip of type k is observed at location x and time i . We can link it to the number of passages recorded with the relationship (1),

$$n(i, x) = \sum_{k=1}^K N_k A_k(i, x) = \langle N, A(i, x) \rangle. \quad (3)$$

The key element of the approach is that the empirical ratio $A_k(i, x)$ is in fact a random perturbation of a *theoretical attendance* $a_k(i, x)$. This function corresponds to patterns in travel behavior between activities. It describes the frequency of visits to a location at given time. For instance, if we consider home-to-work commutes, this ratio will be significantly higher at 8 a.m., when many people are traveling to work, compared to 6 p.m., which is more characteristic of the end of the workday rather than the start. Conversely, for k =work-to-home commutes, the ratio $a_k(i, x)$ is much higher at i =6 p.m. than at i =8 a.m. In other words, $a_k(i, x)$ can be interpreted as the theoretical probability that an individual undertaking the journey type k will pass through location x during time i . In Section 5, we show that the theoretical attendance can be obtained from a microscopic, individual-based stochastic model based on additional geographic data (spatial distribution of activity locations and activity schedules).

Finally, if we have N_k travels related to the activity k , we expect

$$n_k(i, x) \approx N_k a_k(i, x) \quad (4)$$

passages during time interval i through x for the activity k . Across all activities, the linear relationship (1) means that we further expect $\langle N, a(i, x) \rangle$ passages at the same time interval and position.

Our first goal in this article is to disaggregate the hourly traffic counts n by travel purpose, using the approximation

$$n(i, x) \approx \langle N, a(i, x) \rangle.$$

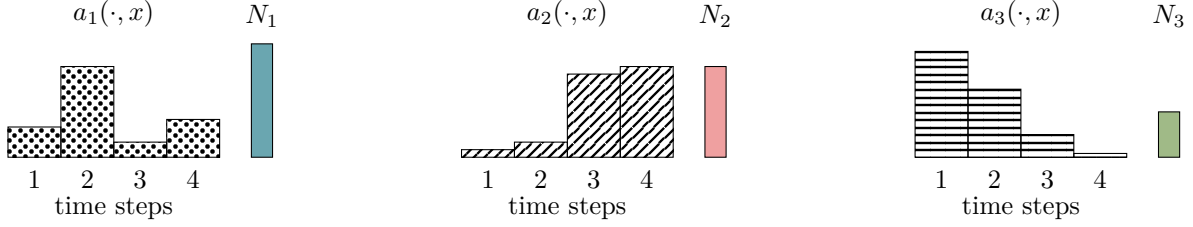
We seek to estimate $\hat{N} \in (\mathbb{R}_+)^K$ which aligns the expected number of trips with the observed data transmitted by the counters. We provide a visual explanation of this challenge in Figure 1.

2.2 Guiding example

The numerical results of the paper will be illustrated through the following guiding example. From some stochastic simulation model, we generate *in silico* the hidden data and vectors of population sizes. From them and (2), we know the ratio defining the empirical attendance functions A_k . In turn, we can define the theoretical attendance function a_k as the expectation of every A_k , i.e. $a_k := \mathbb{E}[A_k]$.

We hereby assume that individuals can engage in three activities: R , L and U (this can be for example staying at home, shopping and working). The spatial distribution of the activities is not homogeneous: people doing L are predominantly located on the left of the spatial domain, people doing R rather on the right and those doing U are uniformly distributed. Activities are conducted following temporal schedules: the end times of the activities are known (for example the end of work schedule). This information is gathered in Figure 2.

Characteristics of journey types: attendance functions and sub-population sizes



Number of passages at location x : expected (and how it is decomposed) vs. data

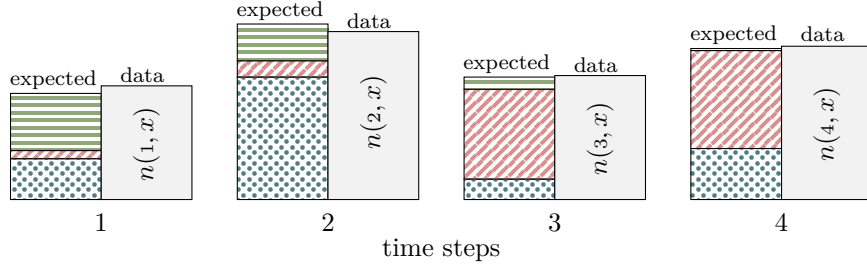


Figure 1: Illustration of the disaggregation challenge with three types of journey ($k = 1, 2, 3$) and four time steps ($i = 1, 2, 3, 4$). On the top, the histograms show theoretical attendance functions (at some counter location x) and sub-population sizes for the three journey types. The bottom diagram shows the scalar product at location x for each time interval i between $(a_k(i, x))_{k \leq 3}$ and $(N_k)_{k \leq 3}$, alongside the observed count data $n(i, x)$ (gray rectangle). The scalar products represent the expected number of passages. Thus, they should closely align with the observed counts $n(i, x)$. Since N is unknown, our strategy is to estimate it by determining the best fit between this expected number of passages and the observed counting data.

For clarity reasons, we only consider four possible types of journey: LR (from L to R), RL , RU and UL . The other types of journey (LU and UR) are assumed to be never observed. We generate data in silico for 300 days, which will be used to build the attendance function. The method for the generation is dictated by the derivation of the empirical function that we propose in Section 5. More precisely, at day $d \leq 300$, we generate $N_k^{(d)}$ trips for a given type of journey $k \in \{LR, RL, RU, UL\}$. We also randomly place 50 counters in the space interval $[0, 1]$, following a uniform distribution.

The generation of trips then includes the associated path and the type of journey. In this simulation scenario, we have therefore access to the hidden data $n_k(i, x_j)$ at each time $i \leq 24$ and position of each counters $(x_j)_{j \leq 50}$, as illustrated in Figure 4.

Given both vector of sub-population sizes by journey and hidden data for 300 different days, we can define, in light of formula (2), (an empirical approximation of) the attendance function $a_k(i, x_j)$, $k \in \{LR, RL, RU, UL\}$, $1 \leq i \leq 24$, $j \leq 50$ as

$$a_k(i, x_j) := \frac{1}{300} \sum_{d=1}^{300} \frac{n_k^{(d)}(i, x_j)}{N_k^{(d)}}. \quad (5)$$

For each activity, we can take a look at the values of a_k , either at a specific counter's location x_j for each hour, as in Figure 5, or at a specific hour across each counter's location, as in Figure 6.

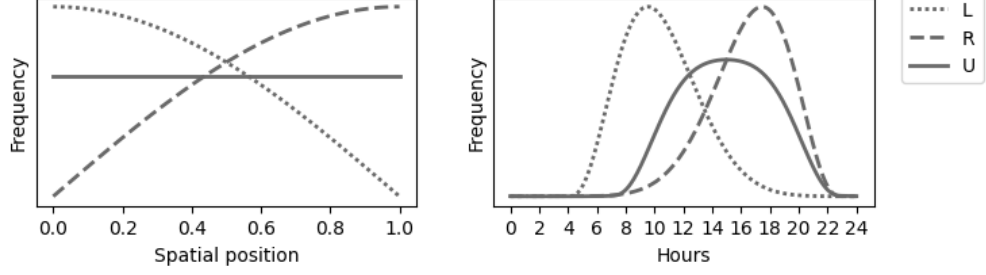


Figure 2: Spatial distribution of activity locations (left) and distribution of end time of each activity (right).

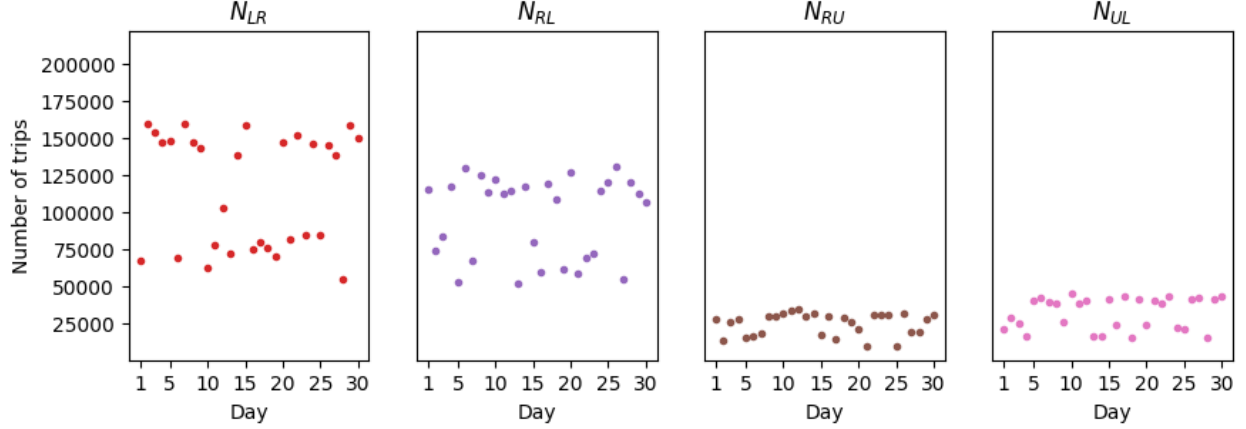


Figure 3: Generated sub-population sizes for the first 30 days (among 300). It can be observed that, for example, LR and RL are much more usual journey types than RU and UL .

3 Statistical model

To estimate the unknown vector of population sizes N , we propose a statistical model, associated with a likelihood function. It depends on the attendance function $a(i, x)$ as well as the count data $n(i, x_j)$, and takes in input a vector of sub-population sizes $\nu \in \mathbb{R}_+^K$. The chosen estimator for N will also be the maximum likelihood estimator (MLE).

The accuracy of this estimation improves as the available counting data grow in size. It is not possible to improve the time resolution of the counters as the counting time intervals (indexed by i) are inherent to the counters. Besides, we would aim to disaggregate the counting data "in real time", *i.e.* over a limited period $i \leq I$.

We thus choose to expand the dataset by increasing the number of counters. We present asymptotic convergence results (law of large numbers, central limit theorem) derived from the statistical model, and we are able to assess the estimation error and evaluate the effectiveness of the chosen spatial arrangement of the counters.

3.1 Statistical hypothesis

Given a journey type $k \leq K$, we model $n_k(i, x)$ (the number of passages in x at time i) as a Poisson random variable of parameter $N_k a_k(i, x)$, introduced as the expected number of passages by x at time step i . This

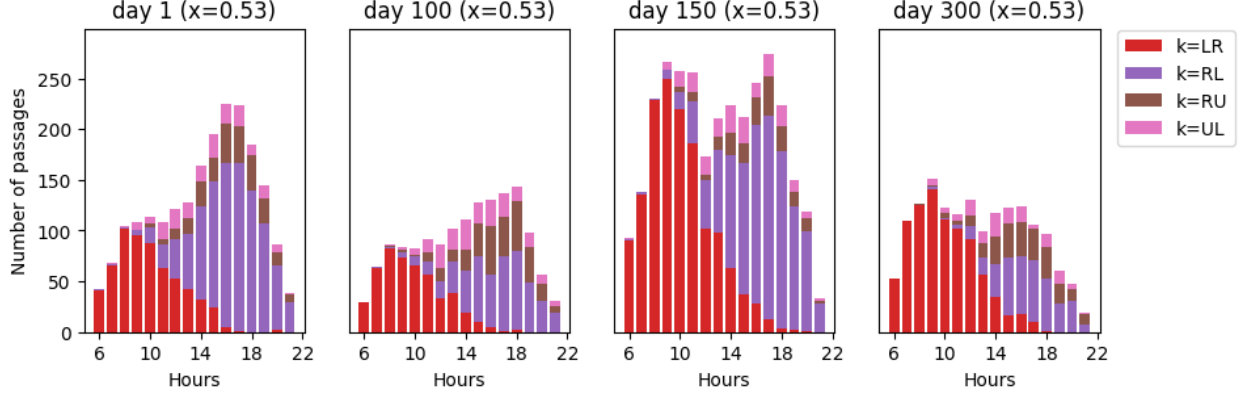


Figure 4: Hidden data for different days, at a single counter location $x_j = 0.53$ and for each hour of the day. The hidden data are stacked by type of journey, and their sum represent the counting data $n(i, x_j)$, pictured by a bullet in Figure 3.

Poisson variable is independent from all other $n_{k'}(i', x')$ (where $(k', i', x') \neq (k, i, x)$).

$$n_k(i, n) \sim \mathcal{P}(N_k a_k(i, x)), \quad \forall i \leq I, \quad \forall x \in \mathcal{E}. \quad (6)$$

Because of the stability property of independent Poisson variables under summation, the aggregated data $n(i, x)$, defined in 1, also follows a Poisson distribution :

$$n(i, x) \sim \mathcal{P}(\langle N, a(i, x) \rangle), \quad \forall i \leq I, \quad x \in \mathcal{E}.$$

These random variable are not identically distributed, as their parameters vary with time interval of counting and with the location of the counters. To achieve a more consistent statistical description of the data, we assume that J counters are positioned randomly in $(x_j)_{j \leq J} \in \mathcal{E}^J$, where each x_j (for $j \leq J$) is drawn independently according to the probability density f_c .

For each counter $j \leq J$, we define the variables y_j , which consist of both position x_j and counts $n(i, x_j)$:

$$y_j := (x_j, (n(i, x_j))_{i \leq I}). \quad (7)$$

Since the locations $(x_j)_{j \leq J}$ are independent and identically distributed, the same holds for the sequence $(y_j)_{j \leq J}$. The randomization of counters position eliminates spatial dependence, while aggregating all time-dependent data per counter removes temporal dependence. The likelihood associated with y_j is derived by conditioning on x_j :

$$\nu \in (\mathbb{R}_+)^K \mapsto \mathcal{L}(\nu, y_j) := f_c(x_j) \prod_{i=1}^I \exp\left(-\langle \nu, a(i, x_j) \rangle\right) \frac{\langle \nu, a(i, x_j) \rangle^{n(i, x_j)}}{n(i, x_j)!}, \quad (8)$$

and the likelihood of the family $(y_j)_{j \leq J}$ is associated to

$$\nu \in (\mathbb{R}_+)^m \mapsto \mathcal{L}(\nu, y_1, \dots, y_J) := \prod_{j=1}^J \mathcal{L}(\nu, y_j). \quad (9)$$

Given the explicit nature of $\mathcal{L}$, the maximum likelihood estimator (MLE) naturally arises as the method of choice for estimating N . This approach is theoretically justified by asymptotic arguments in the following section, after which we propose an efficient algorithm for its computation.

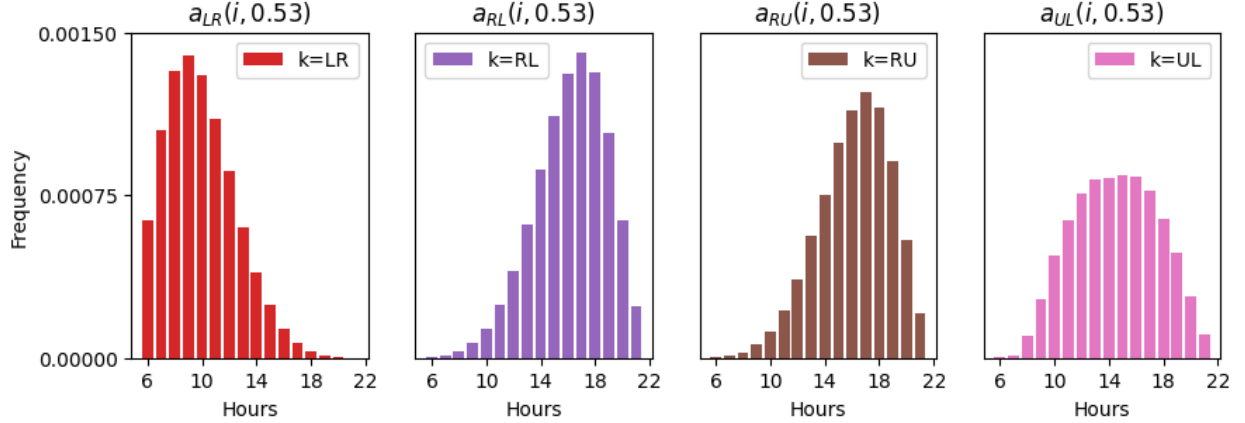


Figure 5: (Empirical approximation of the) attendance function at location $x = 0.53$ for each journey type and each hour of the day, estimated from hidden data of Figure 4. For instance, the attendance function associated to LR (red) tells us that individuals leaving L for R pass through $x = 0.53$ more frequently in the morning than in the evening.

3.2 Maximum-likelihood and asymptotic results in number of counters

We now investigate the asymptotic properties of the MLE as the number of counters J increases. Our goal is to verify that the MLE for this model is consistent and asymptotically normal, mirroring many classical statistical frameworks. More details of this subsection are presented in Appendix A.

Let $(y_j)_{j \in \mathbb{N}^*} = (x_j, (n(i, x_j)))_{j \in \mathbb{N}^*}$ be a sequence of independent and identically distributed random variables, where each y_j follows the distribution $\mathcal{L}(N, \cdot)$ introduced earlier. From now on, we assume that for every $k \leq K$, the attendance function a_k is strictly positive and continuous with respect to its spatial variable (in $\mathcal{E}$). We also assume that the collection of functions $(a_k)_{k \leq K}$ forms a linearly independent family.

The Maximum Likelihood Estimator (MLE) for the observations $(y_j)_{j \leq J}$, denoted by $\hat{N}_J \in (\mathbb{R}_+)^K$ is defined as a maximizer of the likelihood function $\mathcal{L}(\cdot, y_1, \dots, y_J)$, as given in (8) and (9). Note that $\hat{N}_J = 0_K$ if and only if $n(i, x_j) = 0$, for all $i \leq I$ and $j \leq J$. In this case, the likelihood reduces to a Dirac measure at 0_K . Otherwise, the likelihood is strictly positive, smooth, and concave. We want to establish consistency and asymptotic normality of the MLE. We therefore assume that $N \in (\mathbb{R}_+)^K \setminus 0_K$. This guarantees that beyond a certain number of counters J , the MLE $\hat{N}_J$ also lies in $(\mathbb{R}_+)^K \setminus 0_K$.

The concavity of the likelihood associated to this hypothesis as well as the hypothesis over the attendance functions leads to the consistency of the MLE :

$$\hat{N}_J \xrightarrow[J \rightarrow +\infty]{a.e.} N.$$

Before establishing asymptotic normality, we first introduce the Fisher information matrix $\mathcal{I}(N)$, defined as the covariance matrix of the score function (gradient of the log-likelihood). For our model, it admits the following closed-form expression:

$$\mathcal{I}(N) := \mathbb{V}(\log \mathcal{L}(N, y)) = \left(\int_{\mathcal{E}} \frac{a_k(i, x) a_{k'}(i, x)}{\langle N, a(i, x) \rangle} f_c(x) dx \right)_{1 \leq k, k' \leq K}.$$

In particular, $\mathcal{I}(N)$ is invertible. We also deduce the following convergence in distribution:

$$\sqrt{J}(\hat{N}_J - N) \xrightarrow[J \rightarrow +\infty]{\mathcal{L}} \mathcal{N}(0, \mathcal{I}(N)^{-1}).$$

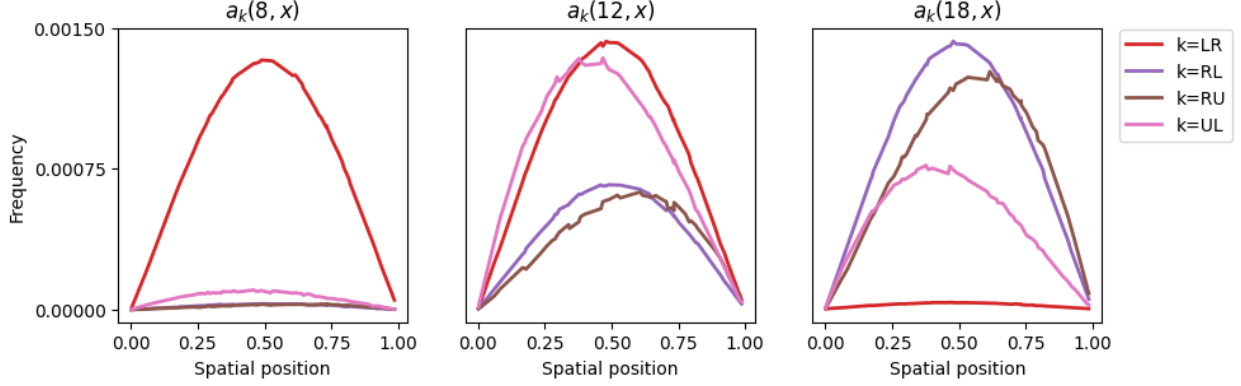


Figure 6: (Empirical approximation of the) attendance as a function of the counter location for each journey type and some hour of the day, estimated from hidden data of Figure 4. For instance, any counter will capture mainly *LR* trips at 8 a.m., but only trips of the three other journey types at 6 p.m.

This central limit theorem classically yields confidence areas for N :

$$\mathbb{P} \left((\hat{N}_J - N)^T \mathcal{I}(N) (\hat{N}_J - N) < \frac{\chi_{K,1-\varepsilon}^2}{J} \right) \xrightarrow{J \rightarrow +\infty} 1 - \varepsilon, \quad (10)$$

where $\left((\hat{N}_J - N)^T \mathcal{I}(N) (\hat{N}_J - N) < \frac{\chi_{K,1-\varepsilon}^2}{J} \right)$ denotes the inner ellipsoid associated to the positive definite matrix $\mathcal{I}(N) / \left(\frac{\chi_{K,1-\varepsilon}^2}{J} \right)$. Inside this ellipsoid, we find N with probability that tends to $1 - \varepsilon$ when $J \rightarrow +\infty$.

As the confidence areas depend on $\mathcal{I}$, and especially on the counter spatial distribution f_c we can try to optimize this distribution in order to improve the accuracy of the estimation with a fixed number of counters, or symmetrically to find the minimum number of counters to put for a specified precision.

4 Approximation of the Maximum Likelihood Estimator

We have established in the Section 3 that the Maximum Likelihood Estimator provides a precise estimate of N , and its Fisher information accurately quantifies the estimation error, depending on the attendance function, but also on the chosen distribution of counters. However, no closed-form expression is available for this estimator. For this reason, we will seek an approximation of the MLE, by adapting the Expectation-Maximisation algorithm to our framework. This will enable us to illustrate the results of Section 3 with our guiding example.

4.1 Adapted version of the Expectation-Maximisation Algorithm

In Section 2, we introduced the hidden data $(n_k(i, x_j))_{i \leq I, j \leq J}$, which denotes the number of passages associated to the activity k at each time interval T_i and counter's location x_j . As a reminder, a counter positioned in x_j cannot observe directly these hidden data, more details in Section 3.1. Instead, during the time interval T_i , the counter records their aggregated sum $n(i, x_j)$, introduced in equation (3). As in equation 7 where we defined the IID family $(y_j)_{j \leq J}$, we define now the family $(y_j^h)_{j \leq J}$ where we replace the count data $n(i, x_j)$ by the hidden data $(n_k(i, x_j))_{k \leq K}$:

$$(y_j^h)_{j \leq J} := \left(x_j, (n_k(i, x_j))_{k \leq K, i \leq I} \right)_{j \leq J} = \left(\text{position of counter, Hidden Data at counter} \right)_{j \leq J}.$$

We assumed in Section 3.1 that these hidden data follow Poisson distributions (see Equation (6)). Therefore, this family is for the same reason as for $(y_j)_{j \leq J}$ an independent and identically distributed family of random variables. It is also associated with an explicit likelihood function. Its formula is provided as in Equations (8) and (9) as follows:

$$\mathcal{L}^h(\cdot, (y_j^h)_{j \leq J}) : \begin{array}{l} (\mathbb{R}_+)^K \longrightarrow \mathbb{R}_+ \\ (\nu_k)_{1 \leq k \leq K} \longmapsto \prod_{k=1}^K \left[\prod_{j=1}^J \left(f_c(x_j) \prod_{i=1}^I \exp(-\nu_k a_k(i, x_j)) \frac{(\nu_k a_k(i, x_j))^{n_k(i, x_j)}}{n_k(i, x_j)!} \right) \right] \end{array} \quad (11)$$

The Expectation-Maximisation (EM) Algorithm [8] approximates the MLE by decomposing the aggregated sum (3), in iterative steps. The following (Theoretical) algorithm is an adapted version of the EM algorithm to our model.

Algorithm 1 Approximation of $\hat{N}_J$: theoretical steps

- 1: initialize $(z_{kij})_{k \leq K}$ randomly in $\mathbb{N}^K$ so that $\sum_{k=1}^K z_{kij} = n(i, x_j)$, $\forall ij$
 - 2: initialize $\nu := \arg \max_{\alpha \in (\mathbb{R}_+)^K} \mathcal{L}^h \left(\alpha, (x_j, (z_{kij})_{k \leq K, i \leq I})_{j \leq J} \right)$
 - 3: **while** ν is evolving **do**
 - 4: $(x_j, (z_{kij})_{k \leq K, i \leq I})_{j \leq J} \leftarrow \mathbb{E}_\nu^h \left[(y_j^h)_{j \leq J} \mid (n(i, x_j))_{i \leq I, j \leq J} \right]$ E-step
 - 5: $\nu \leftarrow \arg \max_{\alpha \in (\mathbb{R}_+)^K} \left(\mathcal{L}^h \left(\alpha, (x_j, (z_{kij})_{k \leq K, i \leq I})_{j \leq J} \right) \right)$ M-step
 - 6: **end while**
 - 7: return $\nu = (\nu_1, \dots, \nu_K)$.
-

In particular, $\mathbb{E}_\nu^h$ (introduced in step 4) represents the conditional expectation of hidden variables under the parameter ν (i.e. with distribution $\mathcal{L}(\nu, \cdot)$).

The strict concavity of the likelihood for a sufficient number of counters J guarantees the convergence of the EM-Algorithm to $\hat{N}_J$, independently of the initial chosen data ([8]). For implementation, we derive explicit expressions for the E and M steps :

- (E-step) The conditionnal expectation under parameter ν of $(y_j^h)_{j \leq J}$ knowing $(y_j)_{j \leq J}$ (also written as $(x_j, (n(i, x_j))_{i \leq I})_{j \leq J}$) is equal to the conditionnal expectation under parameter ν of (y_j^h) knowing $(n(i, x_j))_{i \leq I, j \leq J}$ and is given in closed form by

$$\mathbb{E}_\nu^h \left[(y_j^h)_{j \leq J} \mid (n(i, x_j))_{i \leq I, j \leq J} \right] = \left(x_j, \left(n(i, x_j) \frac{\nu_k a_k(i, x_j)}{\langle \nu, a(i, x_j) \rangle} \right)_{k \leq K, i \leq I} \right)_{j \leq J}.$$

- (M-step) The maximum (hidden) likelihood estimate of $(y_j^h)_{j \leq J} = (x_j, (n_k(i, x_j))_{k \leq K, i \leq I})_{j \leq J}$ is given in closed form by

$$\arg \max_{\nu \in (\mathbb{R}_+)^K} \mathcal{L}^h \left(\nu, (y_j^h)_{j \leq J} \right) = \left(\frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)} \right)_{k \leq K}.$$

(see appendix B for details). We therefore have the two steps of the EM algorithm for computations:

4.2 Guiding example

We come back to the guiding example introduced in Subsection 2.2. We generate counting data on a new day, with a known vector of population sizes $N = (N_{LR}, N_{RL}, N_{RU}, N_{UL}) = (150000, 120000, 30000, 40000)$. The

Algorithm 2 EM algorithm (in practice)

- 1: initialize $(z_{kij})_{k \leq K}$ randomly in $\mathbb{N}^K$ so that $\sum_{k=1}^K z_{kij} = n(i, x_j)$, $\forall ij$
 - 2: initialize ν to $\left(\frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)} \right)_{k \leq K}$
 - 3: **while** ν is evolving **do**
 - 4: $z_{kij} \leftarrow n(i, x_j) \frac{\nu_k a_k(i, x_j)}{\langle \nu, a(i, x_j) \rangle}$, $\forall k \leq K, i \leq I, j \leq J$ E-step
 - 5: $\nu \leftarrow \left(\frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)} \right)_{k \leq K}$ M-step
 - 6: **end while**
 - 7: return ν .
-

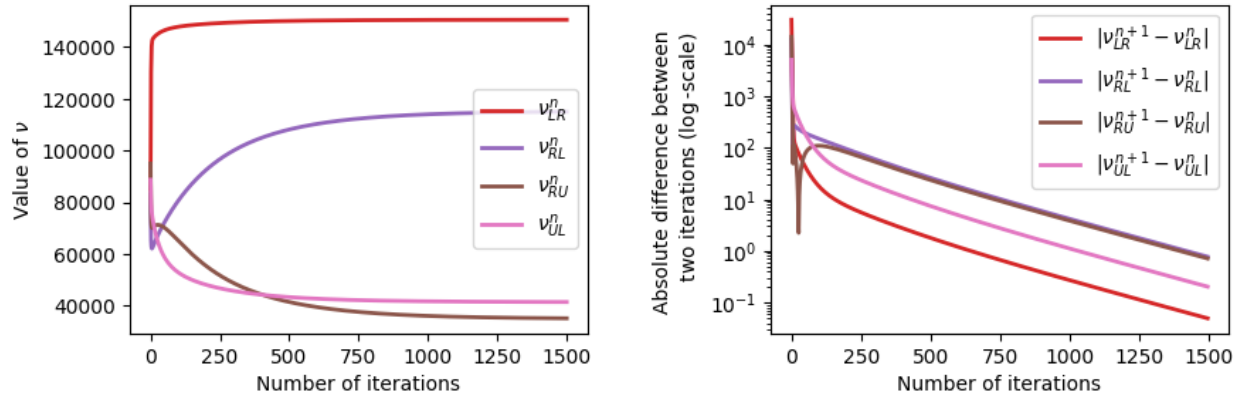


Figure 7: Left: Convergence of the EM-Algorithm toward the Maximum Likelihood Estimate (MLE), right: Absolute difference between two iterations of the EM-algorithm (log-scale), illustrating an exponential convergence

50 counters are still on the same locations as in section 2.2. By using only the counting data $n(i, x_j)_{i \leq 24, j \leq 50}$ and the attendance function defined from prior data by equation (5), we execute the EM-Algorithm to estimate the vector N .

We now illustrate the results of Section 3. For the same vector of population sizes N as in the previous paragraph, we generate trips across fifty days. For each day, we estimate the associated MLE successively with an increasing number of counters, from five up to fifty. The asymptotic confidence intervals given in (10) take the form of a 4-dimensional ellipsoids in this illustrative example. We present 2D slices of such confidence interval, at the 95% confidence level and fifty counters:

Since this confidence interval depends on the distribution of counters, we try out different strategies for positioning the counters. Up to this point, in this guiding example, the counters were distributed uniformly. We now add a second strategy, by placing counters primarily along the boundaries, where one of the activities (L or R) predominates over the other. We also evaluate a third strategy, with counters concentrated on the center of the area, where highest diversity of journey types occurs simultaneously.

More precisely, the three selected distribution over $[0, 1]$ are $d_u : x \mapsto 1$, $d_b : x \mapsto \frac{8}{3} \cos(\pi x)^4$ and $d_c : x \mapsto \frac{8}{3} \sin(\pi x)^4$. Alongside the graph of these distribution, we present an example of 2D slices of the associated confidence ellipsoids for 15 counters, for the three counter distribution strategies.

Up to our knowledge, it is difficult to have an *a priori* intuition on which strategy yields better results. We

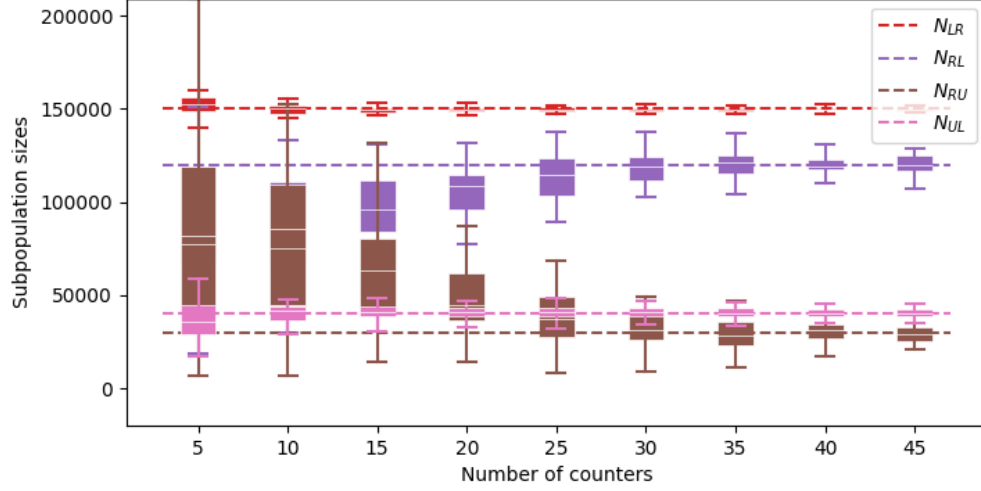


Figure 8: For a fixed number of counters, we plot the boxplots of each coordinate of the MLE derived from 30 distinct datasets. We look at the evolution of these boxplots when the number of counters increases (i.e. when the datasets grow)

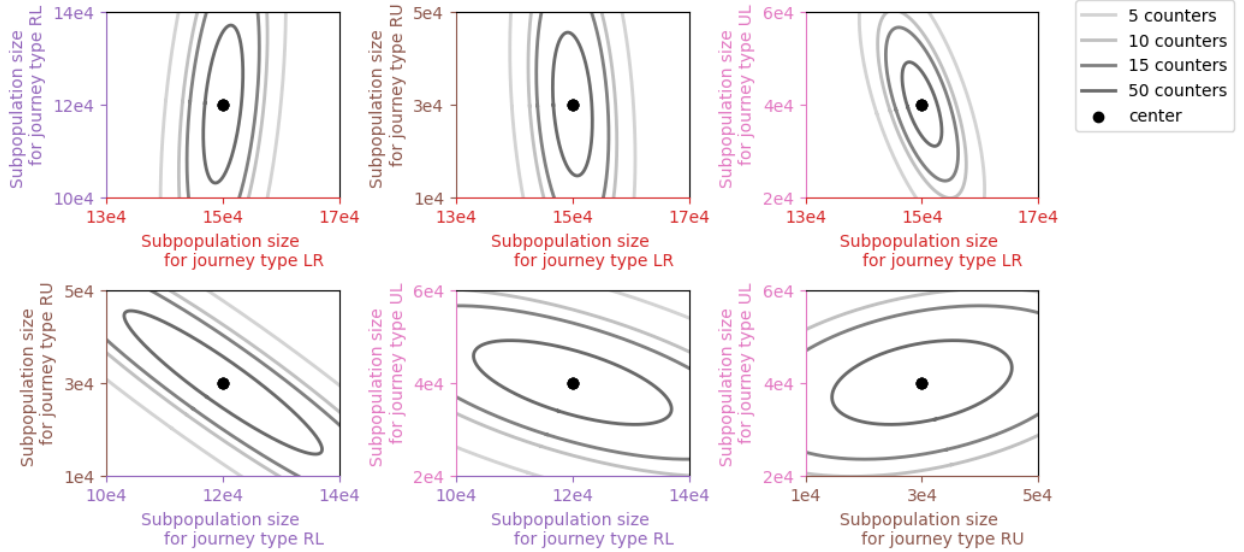


Figure 9: The six different 2D slices of the 4D confidence ellipsoid at 95% rate for the MLE, corresponding to 5, 10, 15, and 50 counters. The lengths of the ellipsoid axes decrease inversely with the square root of the number of counters.

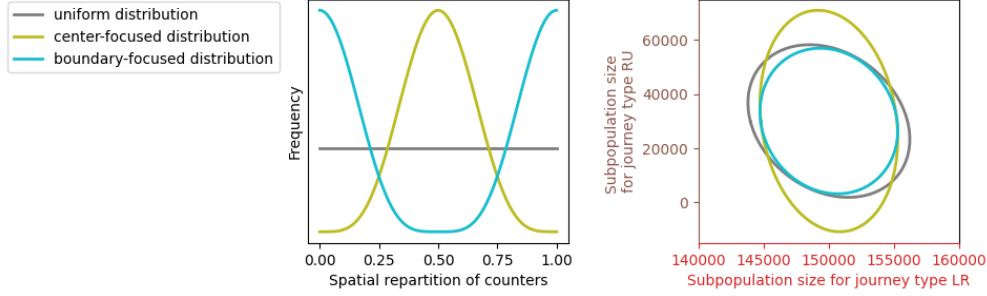


Figure 10: Left: Three distribution strategies with 15 counters over the spatial domain $[0,1]$. Right: one of the six 2D slices (for journey types RU and LR) of the 4D 95% confidence ellipsoid for the MLE, corresponding to each strategy. Among these, the boundary-focused distribution yields the smallest slice area.

therefore use our statistical framework to deduce relevant answers, pictured in Figure 10. The challenge of optimally distributing counters in space reduces to maximizing the Fisher information. It appears to be a promising avenue for developing cost-effective solutions of mobility dynamics analysis.

5 Modelling the attendance function from individual journeys

In this section, we design the attendance function for a given journey-type using socio-geographic static data in a one-dimensional setting. We begin with a stochastic, individual-based model of trips between activities. For large population, we capture a spatio-temporal flux-function. When evaluated at a point x and integrated over the i -th time step, it yields the attendance function $a(i, x)$. In other words, this flux-function can be interpreted as the sum of oriented flux, which are solutions to classical transport equations with source and death terms, analogous to some ecological models. The transport term describes the movement of individuals, while the source term account for individuals leaving their previous activity (*i.e.* entering the study area $\mathcal{E}$). The death term, on the other hand, represent individuals arriving at their next activity (*i.e.* exiting the study area $\mathcal{E}$).

5.1 Individual based model

For a given journey type k , consider N_k individual trajectories over the one dimensional spatial domain $\mathcal{E} = [x_l, x_r]$, and during the time interval $T \subset \mathbb{R}$. This interval is further divided in I disjoint time steps $(T_i)_{i \leq I}$, such that $T = \bigcup_{i \leq I} T_i$. Each individual trajectory, indexed by $p \leq N_k$, describes the continuous, straight-line path of an individual from its origin to its destination. It consists in its position as a function of time. Each trajectory is drawn according to the following random variables:

- v_p , the velocity, with distribution f_v . Each individual travels at a constant velocity, that may vary across individuals. For pedestrians, explicit distributions can be found, *e.g.* in [5].
- x_p^0 , the starting point, with distribution f_{x^0} . It is the location of the previous activity of the individual (*i.e.* before travel). For example, for the journey type "work-to-home commutes", this may represent the density of offices. The cumulative distribution function and survival function of x_p^0 are denoted by F_{x^0} and S_{x^0} . As a reminder, by definition $S_{x^0} := 1 - F_{x^0}$.
- t_p^0 , the starting time, *i.e.* the time at which the individual leaves its previous activity. We further assume that this starting time may also depend on the starting point, x_p^0 . The conditional distribution of starting time has a density (with respect to the Lebesgue measure), $t \mapsto f_{t^0|x^0}(t|x)$, for almost every $x \in \mathcal{E}$. For work-to-home commutes, this may represent the end-of-work schedules at each location. The joint distribution of (t_p^0, x_p^0) is written as $f_{(t^0, x^0)}$.

- $x_p^e \in \mathcal{E}$, the destination point, with distribution f_{x^e} . It is the location of the next activity of the individual. Its distribution function is denoted by F_{x^e} and its survival function by S_{x^e} . For work-to-home commutes, this may represent the density of homes⁴.

We assume v_p , (x_p^0, t_p^0) and x_p^0 are independent random variables. The individual p stands at x_p^0 at time t_p^0 and moves towards x_p^e with velocity v_p . Therefore, at time $t \geq t_p^0$, if it is still moving at this time, its position is given by

$$\gamma_p(t) := x_p^0 + \varepsilon_p(t - t_p^0)v_p, \quad (12)$$

with $\varepsilon_p = 1$ if $x_p^e \geq x_p^0$, -1 else. More precisely, it reaches its destination (to start its new activity) at time $\gamma_p^{-1}(x_p^e) = t_p^0 + \frac{|x_p^e - x_p^0|}{v_p}$. The trajectory of the individual p is thus defined as the restriction of γ_p over the time interval

$$\mathcal{T}_p := \left[t_p^0, t_p^0 + \frac{|x_p^e - x_p^0|}{v_p} \right].$$

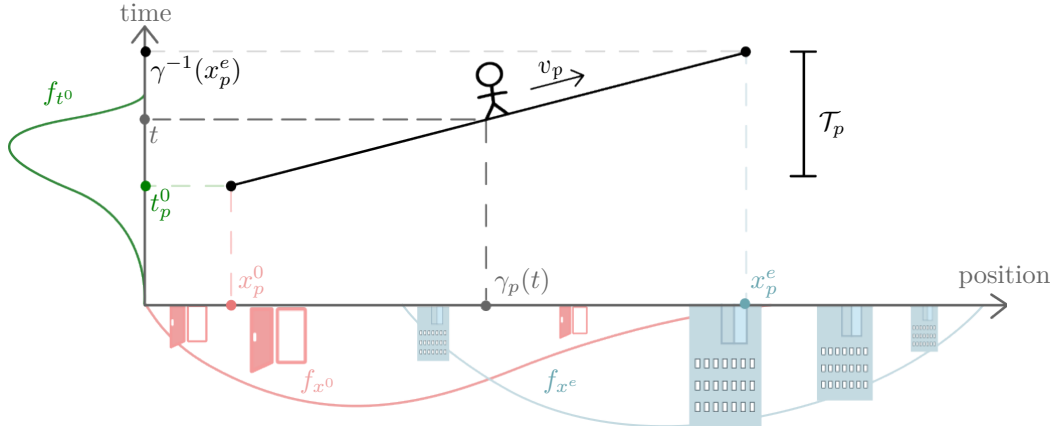


Figure 11: Random trajectory of an individual p . It begins at x_p^0 , drawn from the red distribution (doors) and arrives at x_p^e , drawn from the blue distribution (buildings). The green distribution, used to draw the starting time t_p^0 , depends on x_p^0 . The velocity v_p is also drawn from a distribution f_v (not shown in the graphic). At time $t \in \mathcal{T}_p$, the individual is located at $\gamma_p(t)$.

5.2 Large population model

We still consider a fixed journey type k . Here we show that if the number of (random) individual trips is large, that is $N_k \rightarrow \infty$, then, at any given position x and during any counting interval T_i , ($i \leq I$), the number of passages is deterministic. In other words, with enough trips, we can predict the exact number of people that a counter should be recording.

More precisely, we show that this number can be written as the product $N_k \cdot \int_{t_{i-1}}^{t_i} \nu_k(t, x) dx$. We identify ν_k as a non-oriented flux, that depends only of the expert data f_v , $f_{(t^0, x^0)}$ and f_{x^e} .

Assume that all individuals behave independently from each other. For example, for pedestrians, this is true with a low density of individuals, or anytime that an individual's trajectory is not hindered by surrounding pedestrians. In that case, the family of random variables $(\gamma_p, \mathcal{T}_p)_{d \leq N_k}$ is independent and identically distributed. We consider $n_k(i, x)$ the number of individuals passing through position $x \in \mathcal{E}$ during the time interval T_i :

$$n_k(i, x) := \sum_{p=1}^{N_k} \delta_p, \text{ with } \delta_p = 1 \text{ if } x \in \gamma_p(T_i \cap \mathcal{T}_p), \text{ 0 elsewhere.}$$

⁴https://datafoncier.cerema.fr/sites/datafoncier/files/fichiers/2019/02/CETE_NP_carte_densite_logt_carroyage_100m_A1pdf_0.pdf, last consulted on May 12, 2026.

Define $a_k(i, x) := \mathbb{P}(x \in \gamma(T_i \cap \mathcal{T}_1))$ the probability that a single individual with journey type k passes through x during T_i . The quantity $n_k(i, x)$ is a Bernoulli variable of parameter $(N_k, a_k(i, x))$. The law of large numbers thus guarantees the convergence, almost surely:

$$\frac{1}{N_k} n_k(i, x) \xrightarrow[N_k \rightarrow +\infty]{a.s.} a_k(i, x).$$

In, the Appendix, Section C.1, we show rigorously how this probability $a_k(i, x)$ can be expressed as a the integral of an non-oriented flux function $\nu_k(t, x)$

$$a_k(i, x) = \int_{T_i} \nu_k(t, x) dt. \quad (13)$$

The non-oriented flux $\nu_k(t, x)$ has the following explicit formula, for $t \geq 0$ and $x \in \mathbb{R}$

$$\nu_k(t, x) := \int_{\mathbb{R}_+^*} \int_{\mathbb{R}} G_{0,x}(u) f_{t^0|x^0} \left(t - \frac{|u-x|}{v} \middle| u \right) f_{x^0}(du) f_v(dv). \quad (14)$$

It depends solely on the laws f_v, f_{x^e} and $f_{(t^0, x^0)}$, through their survival and distribution function (denoted by S and F , as in the previous section). The term $\frac{|u-x|}{v}$ represents the remaining travel time from x to u at speed v . Besides, the intermediary function $G_{0,x}$ captures the directionality of travel with respect to x . It is the sum of two terms

$$G_{0,x} : u (\in \mathbb{R}) \mapsto S_{x^e}(x) \mathbb{1}_{]-\infty, x[}(u) + F_{x^e}(x) \mathbb{1}_{[x, +\infty[}(u),$$

Each term of this sum in this definition contributes to one direction of travel, which is clearer once one is reminded that $F_{x^e} = 1 - S_{x^e}$. As a consequence, the flux function itself has two oriented components: a rightward flux $\nu_{\rightarrow}(t, x)$ and a leftward flux $\nu_{\leftarrow}(t, x)$, both of which depend solely on the laws f_v, f_{x^e} and $f_{(t^0, x^0)}$:

$$\nu_{\rightarrow}(t, x) = \int_{\mathbb{R}_+^*} \left[S_{x^e}(x) \int_{-\infty}^x f_{t^0|x^0} \left(t - \frac{x-u}{v} \middle| u \right) f_{x^0}(du) \right] f_v(dv), \quad (15)$$

$$\nu_{\leftarrow}(t, x) = \int_{\mathbb{R}_+^*} \left[F_{x^e}(x) \int_x^{+\infty} f_{t^0|x^0} \left(t - \frac{u-x}{v} \middle| u \right) f_{x^0}(du) \right] f_v(dv). \quad (16)$$

In the rightward flux, we consider the starting times $t - \frac{x-u}{v}$ for an individual passing through x at time t , for any starting position u located to the left of x . Similarly, in the leftward flux, we consider the starting time $t - \frac{u-x}{v}$ for an individual passing through x at time t , for any starting position u located to the right of x .

It is left to the reader to be convinced that the flux ν_k of equation (14) is the the sum of (15) and (16). All the proofs and computation details are given in Section C.1.

5.3 Alternative model with specified arrival time

In the previous individual-based model above, we assumed total knowledge of the distribution of the starting times, for a given journey type. However, depending on the activity, expert data sometimes relates to arrival times rather than starting time (*e.g.* for home-to-work trips). Therefore, when work schedules are used as expert data (see, for example [6]), it becomes necessary to write the model differently. Our approach still works seamlessly in this case, leading once again to explicit formula similar to (14).

We shift slightly the modelling paradigm that lead to (12). First, instead of a starting time, consider an arrival time depending on the arrival position, $x_p^e = x$. The arrival time of an individual p conditionally to $x_p^e = x$ is assumed to admit a density (with respect to the Lebesgue measure) $t(\in \mathcal{E}) \mapsto f_{t^e|x^e}(t|x)$, for almost every $x \in \mathcal{E}$. The joint distribution of (t_p^e, x_p^e) is written as $f_{(t^e, x^e)}$.

This leads to define the analogous trajectory to (12), specific to our new assumptions: for travel time t , individual p locates at position

$$\gamma_p(t) = x_p^e - \varepsilon_p(t_p^e - t)v_p, \quad (17)$$

with $\varepsilon_p = 1$ if $x_p^e \geq x_p^0$, -1 elsewhere. The time interval of the journey is precisely :

$$\mathcal{T} := [\gamma_p^{-1}(x^0), t_p^e] = \left[t_p^e - \frac{|x_p^e - x_p^0|}{v_p}, t_p^e \right].$$

With those notations, for any fixed journey type k , $\frac{1}{N_k}n_k(i, x) \rightarrow a_k(i, x)$, and this attendance function is the sum of two directional flux :

$$a_k(i, x) = \int_{T_i} \nu_{\rightarrow}(t, x)dt + \int_{T_i} \nu_{\leftarrow}(t, x)dt. \quad (18)$$

The difference lies in the definition of the rightward flux $\nu_{\rightarrow}$ and of a leftward flux $\nu_{\leftarrow}$. We show in Section C.1 that $\nu_{\rightarrow}$ is defined as

$$\nu_{\rightarrow}(t, x) := \int_{\mathbb{R}_+^*} \left[F_{x^0}(x) \int_x^{+\infty} f_{t^e|x^e} \left(t + \frac{u-x}{v} \middle| u \right) f_{x^e}(du) \right] f_v(dv), \quad (19)$$

$$\nu_{\leftarrow}(t, x) := \int_{\mathbb{R}_+^*} \left[S_{x^0}(x) \int_{-\infty}^x f_{t^e|x^e} \left(t + \frac{x-u}{v} \middle| u \right) f_{x^e}(du) \right] f_v(dv). \quad (20)$$

The differences between these definitions of oriented fluxes and their analogous with starting time distribution (15) and (16) stem to trajectory considerations: for example, to obtain the rightward flux (19), we consider the arrival time $t + \frac{u-x}{v}$ of an individual passing through x at time t , for any arrival position u located to the right of x . In contrary, in the previous model, we considered the starting time $t - \frac{x-u}{v}$ for an individual passing through x at time t , for any starting position u to the left of x .

5.4 Guiding example

In the guiding example in Section 2.2, for each journey type $k \in \{LR, RL, RU, UL\}$, we defined $a_k(i, x)$ from formula (13), by considering that every individual move with velocity $v = 2$. This amounts to choosing a Dirac velocity distribution $f_V = \delta_2$ in (15). In addition, distributions x^0 , t^0 and x^e as depicted in Figure 2. For example, for journey type $k = LR$, f_{x^0} is the spatial distribution of the activity L (represented by the dotted curve in the left plot of Figure 2), f_{x^e} is the spatial distribution of the activity R (represented by the dashed curve in the left plot of Figure 2) and f_{t^0} is the end time of activity L (represented by the dotted curve in the right plot of Figure 2).

Then, at day d , given the vector of sub-population sizes N^d (see Figure 3), and using approximation (4), we expect $N_k^d a_k(i, x_j)$ passages for journey type k at time-step i and counter location x_j . To generate the synthetic hidden data $n_k^d(i, x_j)$ presented in Figure 4, we added independent Poisson noise to each of these expected passages, following the statistical framework described in Subsection 3.1. Hence, the attendance function that we showed in Figure 5 and 6 are approximate versions of the theoretical attendances from equation (18).

The statistical inference of N via maximum likelihood estimation is performed using these approximate versions of attendance functions in Figure 8. Despite the fluctuation in this data, we successfully achieve precise estimates of the sub-population sizes for a large number of counters.

The attendance functions $a_k(i, x)$ are presented below the same manner as for the approximate attendance functions on Figure 5 and 6: on the fixed position $x = 0.53$ and at every times or at fixed times $t = 8$, $t = 12$, $t = 18$ for every position. 5 and 6 appears to be noised versions of Figures 12 and 13.

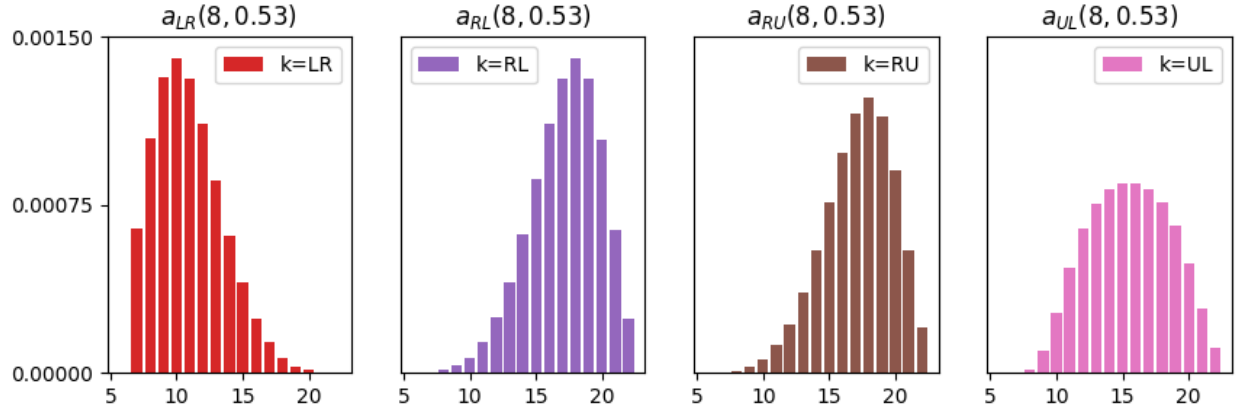


Figure 12: Attendance function computed from the flux formula (14) and distribution function presented in Figure 2, at location $x = 0.53$ at each hour. Figure 5 is an approximate version of this one.

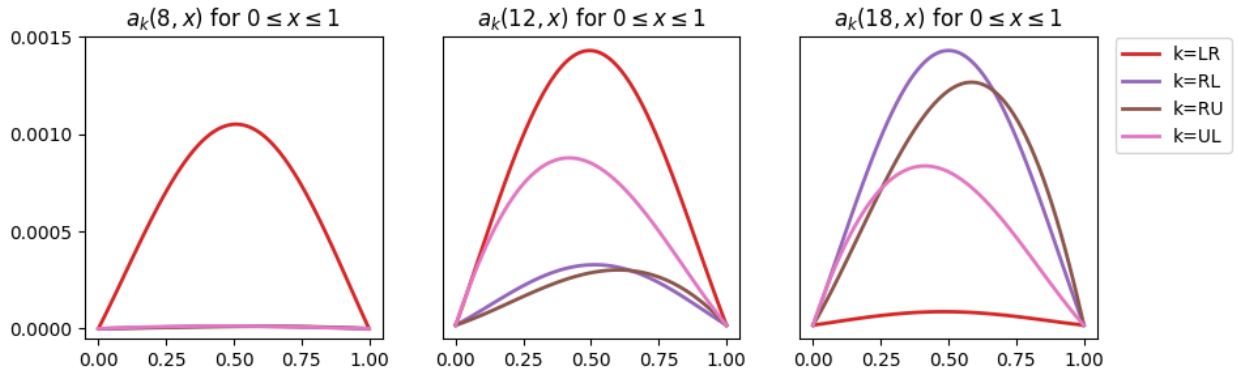


Figure 13: Attendance function computed from the flux formula (14) and distribution function presented in Figure 2, at fixed hours $i = 8$, $i = 12$ and $i = 18$ for every positions. Figure 6 is an approximate version of this one.

5.5 Interpretation of the flux : insight from mathematical biology

If every individual walks at the same speed, we can interpret the flux as the (weak) solution to a transport equation with "source" and "death" terms inspired by mathematical ecology (see for example [16]). In other words, we provide an interpretation of the bracketed terms in the oriented flux expressions (equations (19) and (20), and also their equivalent with arrival time distribution), which means at a given speed, before integration over the velocity distribution. In that spirit, consider a speed v , and the Dirac measure for distribution of speed : δ_v . The flux can be written as

$$\nu_{\rightarrow}(t, x) = S_{x^e}(x) \int_{-\infty}^x f_{t^0|x^0} \left(t - \frac{x-u}{v} \middle| u \right) f_{x^0}(du), \quad (21)$$

$$\nu_{\leftarrow}(t, x) = F_{x^e}(x) \int_x^{+\infty} f_{t^0|x^0} \left(t - \frac{u-x}{v} \middle| u \right) f_{x^0}(du). \quad (22)$$

For the model based on arrival times distribution of Section 5.3, one finds

$$\nu_{\rightarrow}(t, x) = F_{x^0}(x) \int_x^{+\infty} f_{t^e|x^e} \left(t + \frac{u-x}{v} \middle| u \right) f_{x^e}(du), \quad (23)$$

$$\nu_{\leftarrow}(t, x) = S_{x^0}(x) \int_{-\infty}^x f_{t^e|x^e} \left(t + \frac{x-u}{v} \middle| u \right) f_{x^e}(du). \quad (24)$$

We examine the interpretation of the equation (21) in more detail. For the other expressions, the same approach is of course also valid. First, recall that $S_{x^e}(x)$ represents the survival function of the random variable f_{x^e} , which stands for the time of arrival of the individual. By definition, this means that $S_{x^e}(x)$ is the probability that an individual moving to the right will travel to a destination located beyond x (*i.e.* to the right of x).

In the Appendix, Section C.2, we show that the flux $\nu_{\rightarrow}$ defined in (21) is the solution of a transport equation with source and death terms (b and d) of the form

$$\partial_t \nu_{\rightarrow}(t, x) + v \partial_x \nu_{\rightarrow}(t, x) = vb(t, x) - vd(t, x).$$

We recognize a transport equation at speed v , which is consistent with a Dirac speed distribution. Furthermore, we have precise expressions for b and d . The source term is

$$b(t, x) = S_{x^e}(x) f_{(t^0, x^0)}(t, x).$$

This means that new individuals begin a trip rightward, at rate $b(t, x)$. In order to do that, they must follow two conditions. First, they must be at the starting point of the activity at the corresponding time, which happens with probability $f_{(t^0, x^0)}$. In addition, they must begin to travel towards a destination at their right. This is the reason for the appearance of the survival function $S_{x^e}(x)$ here. The individuals starting from point x are partitioned into two groups: those who move left (with proportion $F_{x^e}(x)$) and those who move right (with proportion $S_{x^e}(x)$). This second group is the one that starts their rightward trip, and therefore, enter the count. The other group is accounted for in $\nu_{\leftarrow}$.

The death term is written as

$$d(t, x) = \nu_{\rightarrow}(t, x) \frac{f_{x^e}(x)}{S_{x^e}(x)}. \quad (25)$$

Observe the appearance of the risk function $\frac{f_{x^e}(x)}{S_{x^e}(x)}$ ⁵ in (25) :

$$\frac{f_{x^e}(x)}{S_{x^e}(x)} := \lim_{h \rightarrow 0} \frac{\mathbb{P}(x^e < x + h | x^e \geq x)}{h}.$$

It quantifies the instantaneous risk that the current location x is the arrival location of the trajectory : those individuals are no longer traveling, and therefore exit the count. The death term is also proportional to the flux $\nu_{\rightarrow}$: the risk function is the rate at which individual disappear from the current traveling population.

We have therefore established that $\nu_{\leftarrow}$ verify, in a weak sense, the following transport equation:

$$\partial_t \nu_{\rightarrow}(t, x) + v \partial_x \nu_{\rightarrow}(t, x) = v S_{x^e}(x) f_{(t^0, x^0)}(t, x) - v \nu_{\rightarrow}(t, x) \frac{f_{x^e}(x)}{S_{x^e}(x)}. \quad (26)$$

We slightly abused notation in the right hand side since $f_{(t^0, x^0)}$, and f_{x^e} are not function defined for all (t, x) but probability measures. This explains why this equation is defined on a the dual of $\mathcal{C}_c^\infty$, meaning that for all $\psi \in \mathcal{C}_c^\infty(\mathbb{R} \times \mathbb{R})$:

$$\begin{aligned} & - \int_{\mathbb{R}^2} \nu_{\rightarrow}(t, x) \left[\partial_t \psi(t, x) + v \partial_x \psi(t, x) \right] dt dx \\ & = \int_{\mathbb{R}^2} \psi(t, x) v S_{x^e}(x) f_{(t^0, x^0)}(dt, dx) - \int_{\mathbb{R}} \int_{\mathbb{R}} \psi(t, x) v \nu_{\rightarrow}(t, x) \frac{1}{S_{x^e}(x)} f_{x^e}(dx) dt. \end{aligned}$$

⁵more generally, a risk measure, in the case when f_{x^e} is only a Borel measure, without density with respect to the Lebesgue measure.

This expression is perfectly defined in our functional setting. Note that the left-hand side is the classical weak solution of linear operators, obtained by integration by parts, usually denoted by the duality bracket $\langle \partial_t \nu_{\rightarrow} + v \partial_x \nu_{\rightarrow}, \psi \rangle$. The notation $f_{(t^0, x^0)}(dt, dx)$, $f_{x^e}(dx)$ was used here for was previously denoted as functions in (26). It better conveys the fact that the objects are measures.

Of course, when $f_{(t^0, x^0)}$, f_{x^e} admit a continuous density with respect to the Lebesgue measure, the weak transport equation are well defined in the classical sense.

Similar to this equation, the flux $\nu_{\leftarrow}$ from equation (22) verifies

$$\partial_t \nu_{\leftarrow}(t, x) - v \partial_x \nu_{\leftarrow}(t, x) = v F_{x^e}(x) f_{(t^0, x^0)}(t, x) - v \nu_{\leftarrow}(t, x) \frac{f_{x^e}(x)}{F_{x^e}(x)}.$$

In the case of a specified arrival time instead of starting time (equations (23) and (24)), we have similarly

$$\begin{aligned} \partial_t \nu_{\rightarrow}(t, x) + v \partial_x \nu_{\rightarrow}(t, x) &= v \nu_{\rightarrow}(t, x) \frac{f_{x^0}(x)}{F_{x^0}(x)} - v F_{x^0}(x) f_{(t^e, x^e)}(t, x), \\ \partial_t \nu_{\leftarrow}(t, x) - v \partial_x \nu_{\leftarrow}(t, x) &= v \nu_{\leftarrow}(t, x) \frac{f_{x^0}(x)}{S_{x^0}(x)} - v S_{x^0}(x) f_{(t^e, x^e)}(t, x). \end{aligned} \quad (27)$$

Going back to the formula (15), (16) and their equivalent for arrival times (19) and (20), we interpret the oriented flux as a superposition for all different velocities of transport equations with source and death terms.

Concluding remarks

This work introduces a novel framework for the macroscopic modeling of urban active soft mobility, bridging the gap between aggregated, non-intrusive counting data and the fine-grained characterisation of mobility patterns by trip purpose. To ensure interpretability of the activity-based modelling, we derived our quantity of interest from microscopic, stochastic dynamics. A key contribution lies in the rigorous introduction of attendance functions, which encapsulate the spatio-temporal structure of trips associated with different journey types. This critically enables the disaggregation problem to be reformulated as the estimation of sub-population sizes via standard statistical approach. Consistency and asymptotic normality of the estimator is proven, with, in addition, an adapted EM algorithm to offer a practical and efficient computational solution. Beyond its methodological contributions, our framework highlights the potential of sensor-based data as a robust, privacy-preserving alternative to traditional surveys and digital tracking, as one is able to reconstruct meaningful mobility indicators from widely available urban infrastructures.

This article is a first step opening further academic research in that direction. A natural perspective is the extension of the current one-dimensional modeling to realistic two-dimensional urban networks, where trajectory multiplicity, network topology, and multimodality introduce new theoretical and computational challenges. This transition will require coupling local one-dimensional models along network edges with consistent junction dynamics, as well as integrating mode-dependent route choices. We aim to integrate richer data, such as detailed land-use information and synthetic geo-located agendas (e.g., from projects like ICI, REF a ajouter). This will ensure more realistic attendance functions and thus refine our model. It is also important to notice that it rises significant geovisualization challenges, to account for the joint exploration of mobility intensities and trip purposes of population flows.

Finally, beyond methodological advances, we believe the proposed framework has the potential to become a versatile tool for decision support in sustainable urban mobility. By enabling the reconstruction, interpretation, and comparison of mobility flows by purpose from minimal and non-invasive data, it offers a scalable pathway toward data-informed urban planning, particularly in contexts where traditional or digital mobility data are scarce, biased, or unavailable.

References

- [1] AKANDE, A., CABRAL, P., GOMES, P., AND CASTELEYN, S. The Lisbon ranking for smart sustainable cities in Europe. *Sustainable cities and society* 44 (2019), 475–487.
- [2] BANISTER, D. The sustainable mobility paradigm. *Transport policy* 15, 2 (2008), 73–80.
- [3] BARBAROSSA, L. The post pandemic city: Challenges and opportunities for a non-motorized urban environment. An overview of Italian cases. *Sustainability* 12, 17 (2020), 7172.
- [4] BONNEL, P., HOMBOURGER, E., OLTEANU-RAIMOND, A.-M., AND SMOREDA, Z. Passive mobile phone dataset to construct origin-destination matrix: potentials and limitations. *Transportation Research Procedia* 11 (2015), 381–398.
- [5] CHANDRA, S., AND BHARTI, A. K. Speed distribution curves for pedestrians during walking and crossing. *Procedia-Social and Behavioral Sciences* 104 (2013), 660–667.
- [6] CHENU, A. Les horaires et l’organisation du temps de travail. *Economie et Statistique/Economics and Statistics* 352 (2002), 151–167.
- [7] CIRILLO, C., CORNÉLIS, E., HUBERT, J.-P., AND TOINT, P. Les enquêtes sur les comportements de mobilité, et après ? *Reflets et perspectives de la vie économique* 43, 4 (2004), 111–121.
- [8] DEMPSTER, A. P., LAIRD, N. M., AND RUBIN, D. B. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the royal statistical society: series B (methodological)* 39, 1 (1977), 1–22.
- [9] DEPEAU, S., AND QUESSEVEUR, E. A la recherche d’espaces invisibles de la mobilité. Usages, apports et limites des techniques GPS dans l’étude des déplacements urbains à l’échelle pédestre. *Netcom. Réseaux, communication et territoires*, 28-1/2 (2014), 35–54.
- [10] FEKIH, M., BONNEL, P., SMOREDA, Z., BELLEMANS, T., FURNO, A., AND GALLAND, S. Méthodologie de filtrage et de traitement de données de signalisation de la téléphonie mobile pour la construction de matrice origine-destination: Application à la Région Rhône-Alpes. *Les Cahiers Scientifiques du Transport-Scientific Papers in Transportation* 75 (2019).
- [11] GANAULT, J., AND RAUCH, C. Les enquêtes Emploi du temps: enjeux de mesure et perspectives de recherche. *L’Année sociologique* 252, 2 (2025), 209–231.
- [12] GEBHARDT, L., KRAJZEWICZ, D., OOSTENDORP, R., GOLETZ, M., GREGER, K., KLÖTZKE, M., WAGNER, P., AND HEINRICHS, D. Intermodal urban mobility: users, uses, and use cases. *Transportation Research Procedia* 14 (2016), 1183–1192.
- [13] KEUSCH, F., BÄHR, S., HAAS, G.-C., KREUTER, F., AND TRAPPMANN, M. Coverage error in data collection combining mobile surveys with passive measurement using apps: Data from a German national survey. *Sociological Methods & Research* 52, 2 (2023), 841–878.
- [14] LIU, R., AND LIU, N. Study on urban passenger transport carbon reduction pathways based on system dynamics. *Scientific Reports* 15, 1 (2025), 37471.
- [15] LIVINGSTON, M., MCARTHUR, D., HONG, J., AND ENGLISH, K. Predicting cycling volumes using crowdsourced activity data. *Environment and Planning B: Urban Analytics and City Science* 48, 5 (2021), 1228–1244.
- [16] MURRAY, J. D. *Mathematical biology: I. An introduction*, vol. 17. Springer Science & Business Media, 2007.
- [17] RABAUD, M. EMC²: vers une nouvelle génération d’EMD en France. In *MSFS 2018* (2018).
- [18] RODRIGUEZ-CARRION, A., GARCIA-RUBIO, C., AND CAMPO, C. Detecting and reducing biases in cellular-based mobility data sets. *Entropy* 20, 10 (2018), 736.
- [19] SIŁA-NOWICKA, K., VANDROL, J., OSHAN, T., LONG, J. A., DEMŠAR, U., AND FOTHERINGHAM,

- A. S. Analysis of human mobility patterns from GPS trajectories and contextual information. *International Journal of Geographical Information Science* 30, 5 (2016), 881–906.
- [20] VALLÉE, J., DOUET, A., LE ROUX, G., COMMENGES, H., LECOMTE, C., AND VILLARD, E. Mobiliscope, un outil libre pour explorer les villes et leur mixité sociale au cours des 24 heures de la journée [v4.3], 2024.
- [21] VAN DER VAART, A. W. *Asymptotic statistics*, vol. 3. Cambridge university press, 2000.
- [22] XU, Y., SHAW, S.-L., ZHAO, Z., YIN, L., FANG, Z., AND LI, Q. Understanding aggregate human mobility patterns using passive mobile phone location data: a home-based approach. *Transportation* 42, 4 (2015), 625–646.
- [23] YANG, H., OZBAY, K., AND BARTIN, B. Investigating the performance of automatic counting sensors for pedestrian traffic data collection. In *Proceedings of the 12th world conference on transport research* (2010), vol. 1115, pp. 1–11.

A Asymptotic study of the maximum likelihood estimator

A.1 M-estimation problem

Let first remark that if all the $n(i, x_j)$ are zero,

$$\arg \max_{\nu \in (\mathbb{R}_+)^K} \mathcal{L}(\nu, (y_j)_{j \leq J}) = 0_K.$$

If at least one of the $n(i, x_j)$ is nonzero, then the likelihood is strictly positive for any parameter $\nu \neq 0$ and tends to 0 when $\|\nu\|$ tends to $+\infty$. The MLE is also well defined by continuity and using a compactness argument, but at this point, we do not have unicity. Furthermore, $\nu \mapsto \mathcal{L}(\nu, (y_j)_{j \leq J})$ is strictly positive unless $\nu = 0$. In particular, $\arg \max_{\nu \in (\mathbb{R}_+)^K} \mathcal{L}(\nu, (y_j)_{j \leq J})$ is to be found in $(\mathbb{R}_+)^K \setminus 0$.

We assumed in Subsection 3.2 that $N \neq 0$. With the Kolmogorov’s zero-one law, almost surely from a certain rank J_0 onward, $(n(i, x_j)_{i \leq I})_{j \leq J}$ is not identically zero. In this context, the MLE associated to $(y_j)_{j \leq J}$ is to be found in $(\mathbb{R}_+)^K \setminus 0$, where the log-likelihood is also well-defined. We denote now as $\hat{N}_J$ one MLE associated to $(y_j)_{j \leq J}$. Using the formula of log-likelihood, we can show that $\hat{N}_J$ is an M -estimator [21, Chapter 5]. Indeed:

$$\begin{aligned} \hat{N}_J &= \arg \max_{\nu \in (\mathbb{R}_+)^K \setminus 0} \log \mathcal{L}(\nu, y_1, \dots, y_J) \\ &= \arg \max_{\nu \in (\mathbb{R}_+)^K \setminus 0} \sum_{j=1}^J \log(p(x_j)) + \sum_{j=1}^J \sum_{i=1}^I [-\langle \nu, a(i, x_j) \rangle + n(i, x_j) \log(\langle \nu, a(i, x_j) \rangle) - \log(n(i, x_j)!)] \\ &= \arg \max_{\nu \in (\mathbb{R}_+)^K \setminus 0} \frac{1}{J} \sum_{j=1}^J \sum_{i=1}^I [-\langle \nu, a(i, x_j) \rangle + n(i, x_j) \log(\langle \nu, a(i, x_j) \rangle)] \\ &= \arg \max_{\nu \in (\mathbb{R}_+)^K \setminus 0} \frac{1}{J} \sum_{j=1}^J m(\nu, y_j), \end{aligned}$$

with $m(\nu, y_j) := \sum_{i=1}^I -\langle \nu, a(i, x_j) \rangle + n(i, x_j) \log(\langle \nu, a(i, x_j) \rangle)$. We thus define for all $J \in \mathbb{N}^*$ the M -estimator

$$M_J : \left\{ \begin{array}{ll} (\mathbb{R}_+)^K \setminus 0 & \longrightarrow \mathbb{R} \\ \nu & \longmapsto \frac{1}{J} \sum_{k=1}^J m(\nu, y_k) \end{array} \right.$$

and its associated expectation

$$M : \begin{cases} (\mathbb{R}_+)^K \setminus 0 & \longrightarrow \mathbb{R} \\ \nu & \longmapsto \mathbb{E}[m(\nu, y)] = \mathbb{E} \left[-\langle \nu, a(i, x) \rangle + n_i \log(\nu, a(i, x)) \right] \end{cases}$$

More explicitly, for $\nu \in (\mathbb{R}_+)^K \setminus 0$,

$$\begin{aligned} M(\nu) &= \int_{\mathcal{E}} \sum_{n \in \mathbb{N}^I} \left(\sum_{i=1}^I \left[-\langle \nu, a(i, x) \rangle + n_i \log(\langle \nu, a(i, x) \rangle) \right] \exp(\langle N, a(i, x) \rangle) \frac{\langle N, a(i, x) \rangle^{n_i}}{n_i!} \right) f_c(x) dx \\ &= \sum_{i=1}^I \int_{\mathcal{E}} \left[-\langle \nu, a(i, x) \rangle + \langle N, a(i, x) \rangle \log(\langle \nu, a(i, x) \rangle) \right] f_c(x) dx. \end{aligned}$$

A.2 Consistency of the MLE

We will prove firstly that the MLE is consistent, i.e., $\hat{N}_J \xrightarrow[J \rightarrow +\infty]{\mathbb{P}} N$. This convergence holds even almost surely. We follow the strategy of [21, Chapter 5]. The gradient of $m(\cdot, y)$ (with respect to the first variable) is given for $\nu \in (\mathbb{R}_+)^K \setminus 0$ by

$$\nabla m(\nu, x) = \left(\sum_{i=1}^I a_k(i, x) \left(\frac{n_i}{\langle \nu, a(i, x) \rangle} - 1 \right) \right)_{1 \leq k \leq K}.$$

The gradient of $M(\nu)$ is thus given by

$$\nabla M(\nu) = \nabla \mathbb{E} m(\nu, y) = \mathbb{E} \nabla m(\nu, y) = \sum_{i=1}^I \int_{\mathcal{E}} a_k(i, x) \left(\frac{\langle N, a(i, x) \rangle}{\langle \nu, a(i, x) \rangle} - 1 \right) f_c(x) dx. \quad (28)$$

We also compute the hessian of $m(\cdot, y)$

$$\nabla^2 m(\nu, y) = - \left(\sum_{i=1}^I a_k(i, x) a_{k'}(i, x) \frac{n_i}{\langle \nu, a(i, x) \rangle^2} \right)_{1 \leq k, k' \leq K}$$

and the hessian of M

$$\nabla^2 M(\nu) = \nabla^2 \mathbb{E} m(\nu, y) = \mathbb{E} \nabla^2 m(\nu, y) = - \left(\sum_{i=1}^I \int_{\mathcal{E}} a_k(i, x) a_{k'}(i, x) \left(\frac{\langle N, a(i, x) \rangle}{\langle \nu, a(i, x) \rangle^2} \right) f_c(x) dx \right)_{1 \leq k, k' \leq K}. \quad (29)$$

In particular, $-\nabla^2 M(\nu)$ is the Gram matrix $((a_k | a_{k'}))_{1 \leq k, k' \leq K}$ on the Hilbert space $L^2(\llbracket 1, I \rrbracket \times \mathcal{E})$ with inner product $(\cdot | \cdot)$ defined as

$$(\cdot | \cdot) : \begin{cases} L^2(\llbracket 1, I \rrbracket \times \mathcal{E}) & \times & L^2(\llbracket 1, I \rrbracket \times \mathcal{E}) & \longrightarrow \mathbb{R} \\ (F & , & G) & \longmapsto (F | G) = \sum_{i=1}^I \int_{\mathcal{E}} F(i, x) G(i, x) \left(\frac{\langle N, a(i, x) \rangle}{\langle \nu, a(i, x) \rangle^2} \right) f_c(x) dx \end{cases}$$

It should be noticed that the strict positivity of f_c and of the functions a_k is (almost everywhere) essential to ensure that $\left(\frac{\langle N, a(i, x) \rangle}{\langle \nu, a(i, x) \rangle^2} \right) f_c(x) > 0$ for all $i \leq I$, $x \in \mathcal{E}$, which in turn guarantees that the inner product $(\cdot | \cdot)$ is positive definite.

Since the a_k are assumed to be linearly independent in $\mathcal{C}(\mathcal{E})$, they are also linearly independent in $L^2(\llbracket 1, I \rrbracket \times \mathcal{E})$. We then deduce that $M(\nu)$ is symmetrical positive definite for any $\nu \in (\mathbb{R}_+)^K \setminus 0$. This leads M to be strictly concave over $(\mathbb{R}_+)^K \setminus 0$. In particular, we have the following inequality, $\forall \nu \neq \mu$

$$\langle \nabla M(\nu), \mu - \nu \rangle > M(\mu) - M(\nu). \quad (30)$$

Using the expression of ∇M given in equation (28), we further verify that $\arg \max_{\nu \in (\mathbb{R}_+)^K \setminus 0} M(\nu) = N$. Let $\varepsilon > 0$. By the continuity of M , its image over the sphere centered at N with radius ε , denoted $M(\mathbb{S}(N, \varepsilon))$, is compact. Since N is the unique maximizer of M , it follows from equation (30) that there exists a constant $a > 0$ such that for all $\nu \in \mathbb{S}(N, \varepsilon)$,

$$\langle \nabla M(\nu), N - \nu \rangle > a.$$

Furthermore, as $\nu \in \mathbb{S}(N, \varepsilon) \mapsto m(\nu, y)$ is uniformly continuous for almost all $y \in \mathbb{R} \times \mathbb{N}^I$, the family $(m(\nu, \cdot))_{\nu \in \mathbb{S}(N, \varepsilon)}$ satisfies the Glivenko-Cantelli property [21, Chapter 19]. Consequently, we obtain the almost everywhere uniform convergence over the sphere:

$$\sup_{\nu \in \mathbb{S}(N, \varepsilon)} \|\nabla M_J(\nu) - \nabla M(\nu)\| \xrightarrow[J \rightarrow +\infty]{a.e.} 0.$$

At the same time, we also obtain for any $\nu \neq \mu$

$$\sup_{\nu \in \mathbb{S}(N, \varepsilon)} \|\nabla M_J(\nu) - \nabla M(\nu)\| \xrightarrow[J \rightarrow +\infty]{a.e.} 0.$$

In particular, from some rank J_0 onward, (depending of the sequence $(y_j)_{j \in \mathbb{N}}$), $\langle \nabla M_J(\nu), N - \nu \rangle > a/2$ for all $\nu \in \mathbb{S}(N, \varepsilon)$. This implies that $\hat{N}_J$, the maximizer of M_J lies within the ball $\mathbb{B}(N, \varepsilon)$ almost surely from a certain rank J_0 onward, i.e. for all $J > J_0$,

$$\mathbb{P}\left(\left\|\hat{N}_J - N\right\| > \varepsilon\right) = 0.$$

By applying the Borel-Cantelli lemma, we deduce the almost everywhere convergence:

$$\hat{N}_J \xrightarrow[J \rightarrow +\infty]{a.e.} N. \quad (31)$$

A.3 Asymptotic normality and confidence area

We will now establish that, as in classical settings, consistency is complemented by asymptotic normality. By leveraging the expression in (29) alongside standard results on Fisher information (see e.g. [21]), we obtain a closed form of the Fisher information matrix associated to our statistical model: The Fisher information is defined as

$$\mathcal{I}(N) := \mathbb{V}[\nabla \log \mathcal{L}(N, \cdot)]$$

and is equal to [21, Chapter 5] $\mathcal{I}(N) = \nabla^2 \mathbb{E}[\log \mathcal{L}(N, \cdot)]$, i.e.

$$\begin{aligned} \mathcal{I}(N) &= \mathbb{E}[\nabla^2 \log \mathcal{L}(N, \cdot)] \\ &= \nabla^2 [M(N)] \\ &= \left(\sum_{i=1}^I \int_0^1 \frac{a_k(i, x) a_{k'}(i, x)}{\langle N, a(i, x) \rangle} f_c(x) dx \right)_{1 \leq k, k' \leq K}. \end{aligned}$$

. As discussed earlier, the Fisher information is in particular invertible and leads to the asymptotic normality (for a detailed proof, see [21], Theorem 5.39)

$$\sqrt{J} \left(\hat{N}_J - N \right) \xrightarrow[\mathcal{L}]{J \rightarrow +\infty} \mathcal{N}(0, \mathcal{I}(N)^{-1}).$$

This asymptotic normality enables the construction of confidence regions. By the spectral theorem, we can decompose the Fisher information matrix as $\mathcal{I}(N) = P \Lambda P^T$, where P is an orthogonal matrix and

Λ a diagonal matrix with strictly positive entries (since the Fisher information is positive, definite and symmetric). By stability of convergence in distribution under continuous mapping, we write

$$\begin{aligned}\sqrt{J}\Lambda^{1/2}P^T(\hat{N}_J - N) &\xrightarrow{J \rightarrow +\infty} \Lambda^{1/2}P^T\mathcal{N}_m(0, \mathcal{I}(N)^{-1}), \\ &= \mathcal{N}_m(0, \Lambda^{1/2}P^T\mathcal{I}(N)^{-1}P\Lambda^{1/2}), \\ &= \mathcal{N}_m(0, I_m).\end{aligned}$$

We thus deduce

$$J(\hat{N}_J - N)^T P \Lambda^{1/2} \Lambda^{1/2} P^T (\hat{N}_J - N) \xrightarrow{J \rightarrow +\infty} \chi_K^2,$$

i.e.

$$J(\hat{N}_J - N)^T \mathcal{I}(N) (\hat{N}_J - N) \xrightarrow{J \rightarrow +\infty} \chi_K^2, \quad (32)$$

where χ_K^2 is the chi-squared distribution with K degrees of freedom. We note $\chi_{K,1-\varepsilon}^2$ the associated quantile of order $1 - \varepsilon$. The convergence established in (32) leads finally to the asymptotical confidence area of order $1 - \varepsilon$

$$\mathbb{P} \left((\hat{N}_J - N)^T \mathcal{I}(N) (\hat{N}_J - N) < \frac{\chi_{K,1-\varepsilon}^2}{J} \right) \xrightarrow{J \rightarrow +\infty} 1 - \varepsilon.$$

B Explicit formulas of the EM algorithm

In Section 4, the hidden variables $(n_k(i, x_j))_{k \leq K}$ in each counter position x_j and at each time i are seen as Poisson draws, according to the statistical model that depend on the vector of subpopulation sizes in $\mathbb{R}_+^K \setminus 0$. Recall that this model is associated to the likelihood $\mathcal{L}^h$, introduced in (11). We give details of the computations of the explicit formulas of both Expectation (E) step and Maximisation (M) step.

B.1 E-step

According to Algorithm 1, the E-step consists in computing the conditional expectancy of the hidden data knowing the count data under any vector of subpopulation sizes $\nu \in (\mathbb{R}_+)^K \setminus 0$

$$\mathbb{E}_\nu^h \left[(y_j^h)_{j \leq J} \mid (n(i, x_j))_{i \leq I, j \leq J} \right].$$

We compute for $n \in \mathbb{N}$

$$\begin{aligned}&\mathbb{E}_\nu \left((n_k(i, x_j))_{k \leq K} \mid n(i, x_j) = n \right) \\ &= \sum_{n_1 + \dots + n_K = n} \frac{\mathbb{P}(n_1(i, x_j) = n_1, \dots, n_K(i, x_j) = n_K)}{\mathbb{P}(n(i, x_j) = n)} \cdot \begin{pmatrix} n_1 \\ \vdots \\ n_K \end{pmatrix} \\ &= \sum_{n_1 + \dots + n_K = n} \frac{\prod_{k=1}^K \exp(-\nu_k a_k(i, x_j)) \frac{\nu_k^{n_k} a_k(i, x_j)^{n_k}}{n_k!}}{\exp(-\langle \nu, a(i, x_j) \rangle) \frac{\langle \nu, a(i, x_j) \rangle^n}{n!}} \cdot \begin{pmatrix} n_1 \\ \vdots \\ n_K \end{pmatrix} \\ &= \frac{1}{\langle \nu, a(i, x_j) \rangle^n} \sum_{n_1 + \dots + n_K = n} \binom{n}{n_1, \dots, n_K} \prod_{k=1}^K (\nu_k a_k(i, x_j))^{n_k} \cdot \begin{pmatrix} n_1 \\ \vdots \\ n_K \end{pmatrix},\end{aligned}$$

where $\binom{n}{n_1, \dots, n_K} = \frac{n!}{n_1! \dots n_K!}$. We simplify now each coordinate of the sum: for $q \leq K$,

$$\begin{aligned}
& \sum_{n_1 + \dots + n_K = n} \binom{n}{n_1, \dots, n_K} \prod_{k=1}^K (\nu_k a_k(i, x_j))^{n_k} n_q \\
&= \nu_q a_q(i, x_j) \frac{\partial}{\partial (\nu_q a_q(i, x_j))} \left(\sum_{n_1 + \dots + n_K = n} \binom{n}{n_1, \dots, n_K} \prod_{k=1}^K (\nu_k a_k(i, x_j))^{n_k} \right) \\
&= \nu_q a_q(i, x_j) \frac{\partial}{\partial (\nu_q a_q(i, x_j))} \left(\sum_{k=1}^K \nu_k a_k(i, x_j) \right)^n \\
&= \nu_q a_q(i, x_j) n \langle \nu, a(i, x_j) \rangle^{n-1}.
\end{aligned}$$

We deduce

$$\mathbb{E}_\nu \left((n_k(i, x_j))_{k \leq K} \mid n(i, x_j) \right) = \left(n(i, x_j) \frac{\nu_k a_k(i, x_j)}{\langle \nu, a(i, x_j) \rangle} \right)_{k \leq K}.$$

Since $n_k(i, x_j)$ does not depend on $n(i, x_{j'})$ if $j' \neq j$, we deduce finally that for all $j \leq J$,

$$\mathbb{E}_\nu \left((y_j^h)_{j \leq J} \mid (n(i, x_j))_{i \leq I, j \leq J} \right) = \left(x_j, \left(n(i, x_j) \frac{\nu_k a_k(i, x_j)}{\langle \nu, a(i, x_j) \rangle} \right)_{k \leq K, i \leq I} \right)_{j \leq J},$$

which is the expression of the Expectation step given in Algorithm 2.

B.2 M-step

According to Algorithm 1, the M-step consists in finding the arg max of the hidden likelihood for any hidden data $(z_{kij})_{k \leq K, i \leq I, j \leq J} \in (\mathbb{R}_+)^{KIJ}$

$$\arg \max_{\nu \in (\mathbb{R}_+)^K} \mathcal{L}^h \left(\nu, \left(x_j, (z_{kij})_{k \leq K, i \leq I} \right)_{j \leq J} \right) = \arg \max_{\nu \in (\mathbb{R}_+)^K} \prod_{j=1}^J \mathcal{L}^h \left(\nu, \left(x_j, (z_{kij})_{k \leq K, i \leq I} \right) \right).$$

In the formula (11) of the likelihood with hidden data, for $k \leq K$, ν_k appears only in the k -th product of $\mathcal{L}^h$. Maximizing $\mathcal{L}^h$ is therefore equivalent to maximizing each of the bracketed products in the definition of $\mathcal{L}^h$. For all $y_1^h, \dots, y_J^h = \left(x_j, (n_k(i, x_j))_{i \leq I, k \leq K} \right)_{j \leq J}$ in $(\mathcal{E} \times \mathbb{N}^{I \times K})^J$, we seek

$$\arg \max_{\nu_k \in \mathbb{R}_+} \prod_{j=1}^J \left(p(x_j) \prod_{i=1}^I e^{-a_k(i, x_j) \nu_k} \frac{(a_k(i, x_j) \nu_k)^{n_k(i, x_j)}}{n_k(i, x_j)!} \right).$$

If all $n_k(i, x_j)$ are zero, it is easy to see that the argument of the maximum is 0. Otherwise, we seek an argument of the maximum in $\mathbb{R}_+^*$, where the log-likelihood is well-defined. By the monotonicity of the logarithm, we have

$$\begin{aligned}
& \arg \max_{\nu_k \in \mathbb{R}_+^*} \prod_{j=1}^J \left(p(x_j) \prod_{i=1}^I e^{-a_k(i, x_j) \nu_k} \frac{(a_k(i, x_j) \nu_k)^{n_k(i, x_j)}}{n_k(i, x_j)!} \right) \\
&= \arg \max_{\nu_k \in \mathbb{R}_+^*} \sum_{j=1}^J \left(\log(p(x_j)) + \sum_{i=1}^I -a_k(i, x_j) \nu_k + n_k(i, x_j) \log(a_k(i, x_j) \nu_k) - \log(n_k(i, x_j)!) \right) \\
&= \arg \max_{\nu_k \in \mathbb{R}_+^*} \sum_{i=1}^I \sum_{j=1}^J -a_k(i, x_j) \nu_k + n_k(i, x_j) \log(\nu_k) \\
&= \arg \max_{\nu_k \in \mathbb{R}_+^*} \log(\nu_k) \left(\sum_{i=1}^I \sum_{j=1}^J n_k(i, x_j) \right) - \nu_k \left(\sum_{i=1}^I \sum_{j=1}^J a_k(i, x_j) \right) \\
&= \arg \max_{\nu_k \in \mathbb{R}_+^*} \log(\nu_k) \frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)} - \nu_k \\
&= \frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)}.
\end{aligned}$$

The last line is obtained by studying the function $x \mapsto c \log(x) - x$ with $c > 0$. Thus, since we note that if all $n_k(i, x_j)$ are zero, we can also write $0 = \frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)}$, we deduce

$$\arg \max_{\nu \in (\mathbb{R}_+)^K} (\mathcal{L}(\nu, y_1^h, \dots, y_J^h)) = \left(\frac{\sum_{i,j} n_k(i, x_j)}{\sum_{i,j} a_k(i, x_j)} \right)_{k \in \llbracket 1, K \rrbracket},$$

which is the expression of the Maximisation step given in Algorithm 2.

C Proofs of Section 5 : finding the attendance function and transport equations

C.1 Expression of the attendance function as the integral of a flux

We explain here the formulas of $a_k(i, x)$ in the case of an arrival time distribution, as in section 5.3. The approach is exactly the same in case of a starting time distribution (section 5.1 and section 5.2). All differences between the oriented flux formulas (15), (16), (19) and (20) arise from the version of γ used: this depends on the travel direction (i.e. the choice of $\varepsilon = \pm 1$), and the selected model (i.e., whether γ is expressed in terms of t_{x^e} or t_{x^o}).

We define $(\mathbf{v}, \mathbf{x}^o, \mathbf{t}^e, \mathbf{x}^e, \gamma, \mathcal{T})$ as a random family representing velocity, starting point, ending time, ending point trajectory, and travel interval of an individual with journey type k , with same distribution as any $(v_p, x_p^o, t_p^e, x_p^e, \gamma_p, \mathcal{T}_p)$ for $p \leq N_k$. In particular, $a_k(i, x) = \mathbb{P}(x \in \gamma(T_i \cap \mathcal{T}))$.

We firstly decompose $a(i, x)$ as a rightward and a leftward contributions. This decomposition is achieved by considering the disjoint events $(\mathbf{x}^o \leq \mathbf{x}^e)$ and $(\mathbf{x}^e < \mathbf{x}^o)$, and by using the explicit expression for γ provided

in (17) to compute the probability:

$$\begin{aligned}
\mathbb{P}(x \in \gamma(T_i \cap \mathcal{T})) &= \mathbb{P}\left(\left(\gamma^{-1}(x) \in T_i\right) \cap (x \in \gamma(\mathcal{T}))\right) \\
&= \mathbb{P}\left(\left(\mathbf{t}^e - \frac{\mathbf{x}^e - x}{\mathbf{v}} \in T_i\right) \cap (\mathbf{x}^0 \leq x < \mathbf{x}^e) \cap (\mathbf{x}^0 \leq \mathbf{x}^e)\right) \\
&\quad + \mathbb{P}\left(\left(\mathbf{t}^e - \frac{x - \mathbf{x}^e}{\mathbf{v}} \in T_i\right) \cap (\mathbf{x}^e < x \leq \mathbf{x}^0) \cap (\mathbf{x}^e < \mathbf{x}^0)\right) \\
&= \mathbb{P}\left(\left(\mathbf{t}^e - \frac{\mathbf{x}^e - x}{\mathbf{v}} \in T_i\right) \cap (\mathbf{x}^0 \leq x < \mathbf{x}^e)\right) \\
&\quad + \mathbb{P}\left(\left(\mathbf{t}^e - \frac{x - \mathbf{x}^e}{\mathbf{v}} \in T_i\right) \cap (\mathbf{x}^e < x \leq \mathbf{x}^0)\right).
\end{aligned}$$

By independence of $\mathbf{x}^0$ with other random variables, we have further

$$\begin{aligned}
\mathbb{P}(x \in \gamma(T_i \cap \mathcal{T})) &= F_{x^0}(x) \mathbb{P}\left(\left(\mathbf{t}^e - \frac{\mathbf{x}^e - x}{\mathbf{v}} \in T_i\right) \cap (\mathbf{x}^e - x \geq 0)\right) \\
&\quad + S_{x^0}(x) \mathbb{P}\left(\left(\mathbf{t}^e - \frac{x - \mathbf{x}^e}{\mathbf{v}} \in T_i\right) \cap (x - \mathbf{x}^e \geq 0)\right).
\end{aligned} \tag{33}$$

In the last expression, we decompose $a(i, x)$ into two terms: the probability that an individual arrives at x from the left (i.e. when $\mathbf{x}^0$ is to the left of x), and the probability that an individual arrives at x from the right (i.e. when $\mathbf{x}^0$ is to the right of x). The term $\left(t - \frac{|u-x|}{v}\right)$ represents the time at which an individual, traveling at speed v , passes through x before reaching u at time t (assuming x lies on their trajectory). We now express these two contributions in integral form. To do so, we use the conditional density of $\mathbf{t}^e$ knowing $\mathbf{x}^e = u$, leveraging the independence between $(\mathbf{t}^e, \mathbf{x}^e)$ and $\mathbf{v}$:

$$\begin{aligned}
a(i, x) &= F_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{1}_{(t-(u-x)/v \in T_i)} \mathbb{1}_{((u-x) \geq 0)} f_{t^e|x^e}(t|u) dt \right) f_{x^e}(du) \right) f_v(dv) \\
&\quad + S_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbb{1}_{(t-(x-u)/v \in T_i)} \mathbb{1}_{((x-u) \geq 0)} f_{t^e|x^e}(t|u) dt \right) f_{x^e}(du) \right) f_v(dv) \\
&= F_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_x^{+\infty} \left(\int_{\mathbb{R}} \mathbb{1}_{(t-(u-x)/v \in T_i)} f_{t^e|x^e}(t, u) dt \right) f_{x^e}(du) \right) f_v(dv) \\
&\quad + S_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_{-\infty}^x \left(\int_{\mathbb{R}} \mathbb{1}_{(t-(x-u)/v \in T_i)} f_{t^e|x^e}(t|u) dt \right) f_{x^e}(du) \right) f_v(dv) \\
&= F_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_x^{+\infty} \left(\int_{T_i} f_{x^e|y^e} \left(t + \frac{x-u}{v} \middle| u \right) dt \right) f_{x^e}(du) \right) f_v(dv) \\
&\quad + S_{x^0}(x) \int_{\mathbb{R}_+^*} \left(\int_{-\infty}^x \left(\int_{T_i} f_{x^e|t^e} \left(t + \frac{u-x}{v} \middle| u \right) dt \right) f_{x^e}(du) \right) f_v(dv).
\end{aligned}$$

Finally by the Fubini Theorem:

$$\begin{aligned}
a(i, x) &= F_{x^0}(x) \int_{T_i} \int_x^{+\infty} \int_{\mathbb{R}_+^*} f_{t^e|x^e} \left(t + \frac{u-x}{v} \middle| u \right) f_v(dv) f_{x^e}(du) dt \\
&\quad + S_{x^0}(x) \int_{T_i} \int_{-\infty}^x \int_{\mathbb{R}_+^*} f_{t^e|x^e} \left(t + \frac{x-u}{v} \middle| u \right) f_{x^e}(du) dt,
\end{aligned}$$

which gives the form

$$a(i, x) = \int_{T_i} \nu_{\rightarrow}(t, x) + \int_{T_i} \nu_{\leftarrow}(t, x).$$

Equivalently, if we want a more compact form, we write from last equation

$$a(i, x) = \int_{T_i} \int_{\mathbb{R}_+^*} \int_{\mathbb{R}} G_{0,x}(u) f_{t^e|x^e} \left(t + \frac{|u-x|}{v} \middle| u \right),$$

where we recall that $G_{0,x} := S(x)\mathbb{1}_{]-\infty, x]} + F_{x^0}(x)\mathbb{1}_{]x, +\infty[}$.

C.2 Derivation of the transport equations

We provide a detailed derivation of the transport equation for the rightward flux in the case of an ending time distribution, as defined in (23). The other cases follow analogously.

Step 1: Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^2, \mathbb{R})$. To derive the weak form of the partial differential equation in (27), we test it against ψ . For $y \in \mathcal{E}$, we define the trajectory $g_y : t \in \mathbb{R} \mapsto y + vt$ and its graph $\Gamma_y : t \in \mathbb{R} \mapsto (t, g_y(t))$. Since g_y is invertible for all $y \in \mathcal{E}$, with inverse given by $g_x^{-1}(u) = \frac{u-x}{v}$ for all $u \in \mathbb{R}$, we write by change of variable $x = g_y(t)$, $dx = dy$:

$$\begin{aligned} \langle \partial_t \nu_{\rightarrow} + v \partial_x \nu_{\rightarrow}, \psi \rangle &= \int_{\mathbb{R}} \int_{\mathbb{R}} \nu_{\rightarrow}(t, x) (-\partial_t \psi(t, x) - v \partial_x \psi(t, x)) dx dt \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \nu_{\rightarrow}(t, y + vt) (-\partial_t \psi(t, y + vt) - v \partial_x \psi(t, y + vt)) dy dt \\ &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} -\nu_{\rightarrow} \circ \Gamma_y(t) (\psi \circ \Gamma_y)'(t) dt \right] dy \\ &= \int_{\mathbb{R}} \langle (\nu_{\rightarrow} \circ \Gamma_y)', \psi \circ \Gamma_y \rangle dy. \end{aligned} \tag{34}$$

Step 2: Let $x \in \mathbb{R}$. Let $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}, \mathbb{R})$ a test function on $\mathbb{R}$. We have

$$\begin{aligned} \langle (\nu_{\rightarrow} \circ \Gamma_x)', \varphi \rangle &= \langle \nu_{\rightarrow} \circ \Gamma_x, -\varphi' \rangle \\ &= \int_{\mathbb{R}} -\nu_{\rightarrow}(t, g_x(t)) \varphi'(t) dt \\ &= \int_{\mathbb{R}} -F_{x^0}(g_x(t)) \int_{g_x(t)}^{+\infty} f_{t^e|x^e} \left(\frac{u-x}{v} \middle| u \right) f_{x^e}(du) \varphi'(t) dt, \end{aligned}$$

We observe that $\frac{u-x}{v} = g_x^{-1}(u)$. By applying Fubini's theorem and using the equivalence $u \geq g_x(t) \Leftrightarrow t \leq g_x^{-1}(u)$, we obtain:

$$\langle (\nu_{\rightarrow} \circ \Gamma_x)', \varphi \rangle = \int_{\mathbb{R}} \left(\int_{-\infty}^{g_x^{-1}(u)} -F_{x^0}(g_x(t)) \varphi'(t) dt \right) f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du). \tag{35}$$

The inner integral can be rewritten, first through the change of variable $t = g_x^{-1}(a)$, and then considering integration by part (in the Stieltjes sense):

$$\begin{aligned} \int_{-\infty}^{g_x^{-1}(u)} -F_{x^0}(g_x(t)) \varphi'(t) dt &= \int_{-\infty}^u -F_{x^0}(a) (\varphi \circ g_x^{-1})'(a) da \\ &= [-(\varphi \circ g_x^{-1}) F_{x^0}]_{-\infty}^u + \int_{-\infty}^u \varphi(g_x^{-1}(a)) f_{x^0}(da) \\ &= -\varphi(g_x^{-1}(u)) F_{x^0}(u) + \int_{-\infty}^u \varphi(g_x^{-1}(a)) f_{x^0}(da). \end{aligned} \tag{36}$$

We finally plug (36) into (35) to deduce:

$$\begin{aligned}
\langle (\nu_{\rightarrow} \circ \Gamma_x)' | \varphi \rangle &= \int_{\mathbb{R}} -\varphi(g_x^{-1}(u)) F_{x^0}(u) f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du) \\
&\quad + \int_{\mathbb{R}} \left(\int_{-\infty}^u \varphi(g_x^{-1}(a)) f_{x^0}(da) \right) f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du) \\
&= \int_{\mathbb{R}} -\varphi(g_x^{-1}(u)) F_{x^0}(u) f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du) \\
&\quad + \int_{\mathbb{R}} \left(\int_a^{+\infty} f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du) \right) \varphi(g_x^{-1}(a)) f_{x^0}(da) \\
&= \int_{\mathbb{R}} -\varphi(g_x^{-1}(u)) F_{x^0}(u) f_{t^e|x^e}(g_x^{-1}(u)|u) f_{x^e}(du) \\
&\quad + \int_{\mathbb{R}} \frac{1}{F_{x^0}(a)} \nu_{\rightarrow}(g_y^{-1}(a), a) \varphi(g_x^{-1}(a)) f_{x^0}(da).
\end{aligned} \tag{37}$$

Step 3: We plug (37) into (34) to obtain:

$$\begin{aligned}
\langle \partial_t \nu_{\rightarrow} + v \partial_x \nu_{\rightarrow}, \psi \rangle &= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} -\psi(g_y^{-1}(u), u) F_{x^0}(u) f_{t^e|x^e}(g_y^{-1}(u)|u) f_{x^e}(du) \right] dy \\
&\quad + \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \frac{1}{F_{x^0}(a)} \nu_{\rightarrow}(g_y^{-1}(a), a) \psi(g_y^{-1}(a), a) f_{x^0}(da) \right] dy \\
&= \int_{\mathbb{R}} \left[\int_{\mathbb{R}} -\psi\left(\frac{u-y}{v}, u\right) F_{x^0}(u) f_{t^e|x^e}\left(\frac{u-y}{v} | u\right) dy \right] f_{x^e}(du) \\
&\quad + \int_{\mathbb{R}} \left[\int_{\mathbb{R}} \frac{1}{F_{x^0}(a)} \nu_{\rightarrow}\left(\frac{a-y}{v}, a\right) \psi\left(\frac{a-y}{v}, a\right) dy \right] f_{x^0}(da).
\end{aligned}$$

Finally, by change of variable $y = u - vt$, $dy = -vdt$ and $y = a - vt$, $dy = -vdt$, we deduce

$$\begin{aligned}
\langle \partial_t \nu_{\rightarrow} + v \partial_x \nu_{\rightarrow}, \psi \rangle &= \int_{\mathbb{R}} \int_{\mathbb{R}} -\psi(t, u) F_{x^0}(u) f_{t^e|x^e}(t|u) v dt f_{x^e}(du) \\
&\quad + \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{1}{F_{x^0}(a)} \nu_{\rightarrow}(t, a) \psi(t, a) v dt f_{x^0}(da) \\
&= \int_{\mathbb{R}^2} -\psi(t, u) v F_{x^0}(u) f_{(t^e, x^e)}(dt, du) + \int_{\mathbb{R}} \int_{\mathbb{R}} \psi(t, a) v \nu_{\rightarrow}(t, a) \frac{1}{F_{x^0}(a)} f_{x^0}(da) dt \\
&= \langle -v F_{x^0}(x) f_{(t^e, x^e)}(dt, dx) + v \nu_{\rightarrow}(t, x) \frac{1}{F_{x^0}(x)} f_{x^0}(dx), \psi \rangle.
\end{aligned}$$

i.e.

$$\partial_t \nu_{\rightarrow}(t, x) + v \partial_x \nu_{\rightarrow}(t, x) = -v F_{x^0}(x) f_{(t^e, x^e)}(t, x) + v \nu_{\rightarrow}(t, x) \frac{f_{x^0}(x)}{F_{x^0}(x)},$$

in the weak sense, on the dual of the space of test functions $\mathcal{C}_c^{\infty}$. This indeed recovers Equation (27).